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Controls on the breach geometry and flood hydrograph during overtopping of noncohesive earthen dams

Joseph S. Walder[1], Richard M. Iverson[1], Jonathan W. Godt[2], Matthew Logan[1], and Stephen A. Solovitz[3]

[1]U.S. Geological Survey, Cascades Volcano Observatory, Vancouver, Washington, USA, [2]U.S. Geological Survey, Golden, Colorado, USA, [3]School of Engineering and Computer Science, Washington State University—Vancouver, Vancouver, Washington, USA

Key Points:
- Hydrographs are reproducible after early stage affected by groundwater/surface-water interaction
- Erosional step-forming instability is key process that triggers onset of rapid reservoir drainage
- Measured shape of the hydraulic-control cross section differs from common model assumption

Supporting Information:
- Movie S1
- Movie S2
- Movie S3
- Movie S4
- Movie S5
- Movie S6
- Movie S7
- Movie S8
- Movie S9
- Data Sets S1, S3, S4, S6, S7, S9, S11–S15, and S20–S24 and Tables S1–S4
- Data Sets S2, S5, S8, S10, and S16–S19

Correspondence to:
J. S. Walder,
jswalder@usgs.gov

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Abstract Overtopping failure of noncohesive earthen dams was investigated in 13 large-scale experiments with dams built of compacted, damp, fine-grained sand. Breaching was initiated by cutting a notch across the dam crest and allowing water escaping from a finite upstream reservoir to form its own channel. The channel developed a stepped profile, and upstream migration of the steps, which coalesced into a headcut, led to the establishment of hydraulic control (critical flow) at the channel head, or *breach crest*, an arcuate erosional feature that functions hydraulically as a weir. Novel photogrammetric methods, along with underwater videography, revealed that the retreating headcut maintained a slope near the angle of friction of the sand, while the cross section at the breach crest maintained a geometrically similar shape through time. That cross-sectional shape was nearly unaffected by slope failures, contrary to the assumption in many models of dam breaching. Flood hydrographs were quite reproducible—for sets of dams ranging in height from 0.55 m to 0.98 m—when the time datum was chosen as the time that the migrating headcut intersected the breach crest. Peak discharge increased almost linearly as a function of initial dam height. Early-time variability between flood hydrographs for nominally identical dams is probably a reflection of subtle experiment-to-experiment differences in groundwater hydrology and the interaction between surface water and groundwater.

1. Introduction

Floods from earthen-dam failures have long been recognized not only for their destructiveness but also as agents of rapid, large-scale geomorphic change [*O'Connor and Beebee*, 2009]. Reliable and accurate mathematical methods to model dam-breach evolution have therefore been sought by engineers and hydrologists who need to delineate flood hazards zones, as well as by geologists interested in reconstructing paleofloods. Two quite distinct modeling approaches have developed. The approach that most completely incorporates the relevant physics involves computationally solving coupled differential equations for fluid- and sediment transport over an irregular, erodible bed. The equations are customarily depth-averaged and solved over a two-dimensional domain [e.g., *Wu et al*., 2012]. In a morphodynamic model of this sort, the evolving dam breach is simply part of the overall computational domain and has no special properties. Alternatively, in a "parametric" modeling approach such as that of *Wu* [2013], the flood flow is envisaged as being routed through a prismatic channel—in effect, a gutter—incised into the dam crest and the downstream face of the dam (Figure 1). Flow control associated with the transition from subcritical to supercritical flow, as water escapes from a reservoir, is also handled differently: in a morphodynamic model, flow transitions arise naturally as part of the solution, whereas in existing parametric models, the flow transition is assumed to occur somewhere along the gutter, but the exact location of the flow transition is unimportant. Hydraulically, the gutter is treated as a broad-crested weir, with the discharge Q depending on the size and shape of the control section. In the earliest parametric models, the cross-sectional shape and rate of downcutting were specified [e.g., *Fread*, 1977]. In later parametric models, the breach was still treated as a gutter, but the cross-sectional shape everywhere, including at the flow transition, was assumed to be trapezoidal and to reflect a competition between slumping, which widens the breach channel, and fluvial sediment transport, which removes the slumped sediments [e.g., *Fread*, 1985]. Some version of this hypothesis informs all parametric models of dam breaching, including recent ones [e.g., *Macchione*, 2008; *Wu*, 2013]. However, it is inconsistent with the findings we report here.

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《水资源研究》

研究论文
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非黏性土坝漫顶失事过程中溃口几何形态与洪水过程线的控制因素

约瑟夫·S·沃尔德[1]，理查德·M·伊弗森[1]，乔纳森·W·戈德特[2]，马修·洛根[1]，斯蒂芬·A·索洛维茨[3]

1美国地质调查局，喀斯喀特火山观测站，华盛顿州温哥华市，美国；2美国地质调查局，科罗拉多州戈尔登市，美国；3华盛顿州立大学—温哥华分校，工程与计算机科学学院，华盛顿州温哥华市，美国

关键要点：
• 洪水过程线在受地下水/地表水相互作用影响的早期阶段之后具有可重复性
• 侵蚀阶地形成失稳是触发水库快速排水的关键过程

• 实测水力控制断面形状与常见模型假定不符

支持信息：
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• 数据集S2、S5、S8、S10及S16–S19

通讯作者：
J. S. 沃尔德，
jswalder@usgs.gov

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摘要：本研究通过13组大型实验，考察了非黏性土坝漫顶失事过程，实验所用坝体由压实、含水的细粒砂筑成。溃坝起始于在坝顶开凿缺口，使有限上游水库中的水自行冲刷形成通道。该通道发育为阶梯状剖面，阶梯向上游迁移并融合为溯源侵蚀坎，最终在通道顶端（即溃口顶部）建立水力控制（临界流）。溃口顶部是一种弧形侵蚀特征，其水力学功能相当于堰。新颖的摄影测量方法结合水下视频技术揭示：后退的溯源侵蚀坎坡度始终接近砂体的内摩擦角，而溃口顶部的横断面形状在时间进程中保持几何相似性。该横断面形状几乎不受边坡失稳影响，这与众多溃坝模型的假定恰恰相反。对于坝高介于0.55米至0.98米之间的一系列坝体，若以溯源侵蚀坎与溃口顶部交汇时刻为时间零点，则所获洪水过程线具有高度可重复性。峰值流量几乎随初始坝高呈线性增长。名义上完全相同的坝体在早期阶段洪水过程线间的差异，可能反映了各次实验中地下水水文状况及地表水—地下水相互作用的细微差别。

1．引言

土坝失事引发的洪水，长期以来不仅因其破坏性而受到关注，亦被视为快速、大规模地貌变化的重要营力[奥康纳与比比，2009]。因此，亟需可靠而精确的数学方法来模拟溃坝演化过程，以便工程师与水文学家划定洪水危险区，以及地质学家重建古洪水事件。目前已形成两种截然不同的建模方法。其中一种方法最完整地纳入了相关物理机制，即在不规则、可侵蚀床面上联立求解流体与泥沙输运的微分方程组。此类方程通常采用水深平均形式，并在二维区域上求解[例如，吴等，2012]。在此类地貌动力学模型中，不断演化的溃口仅是整体计算域的一部分，不具特殊属性。另一种方法为参数化建模法，如吴[2013]所采用的方法：将洪水过程设想为经一棱柱形渠道（实质上是一条沟槽）泄流，该沟槽切入坝顶及坝体下游坡面（图1）。至于水流由亚临界流向超临界流的过渡所关联的水力控制问题，两种建模方法处理方式亦有差异：在地貌动力学模型中，流态过渡自然成为解的一部分；而在现有参数化模型中，流态过渡被假定发生在沟槽某处，但其确切位置并不重要。从水力学角度看，该沟槽被当作宽顶堰处理，其泄流量Q取决于控制断面的尺寸与形状。在最早的参数化模型中，横断面形状与下切速率均为给定值[例如，弗雷德，1977]。在后续参数化模型中，溃口仍被视作沟槽，但其全断面（包括流态过渡断面）形状均被假定为梯形，以反映滑塌作用（拓宽溃口通道）与河流泥沙输运（带走滑塌沉积物）之间的竞争关系[例如，弗雷德，1985]。此假说的某种变体构成了所有溃坝参数化模型（包括近期模型[例如，马基奥内，2008；吴，2013]）的基础。然而，它与本文报告的发现相矛盾。

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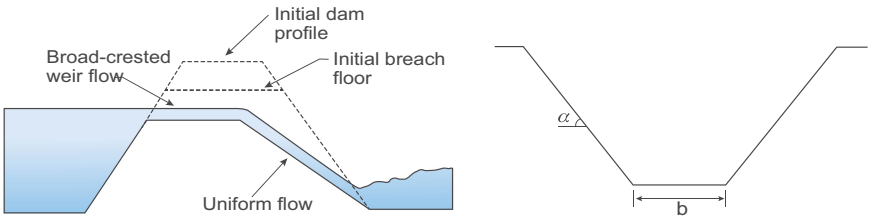


Figure 1. Breach geometry commonly but inappropriately assumed in parametric models [after *Wu et al*., 2009, by permission of the International Association for Hydro-Environment Engineering and Research]. Water flows in a gutter that traverses first the eroding dam crest (where hydraulic control is assumed to exist) and then the downstream dam face, at the base of which a hydraulic jump occurs. The cross sectional shape of the gutter (at right) is commonly taken as trapezoidal, with bottom width b and side slope α specified.

The utility of parametric dam-breach models lies primarily in the fact that, with much less computational effort and expense than morphodynamic models, they can provide insight into the way that a computed flood hydrograph is affected by diverse parameters that characterize breach evolution. This is only possible, of course, if a parametric model represents the physical system correctly. However—and here we anticipate results presented in more detail below—when one observes an actual dam-breaching event instead of just its aftermath, it becomes apparent that little water drains during the period in which there is a gutter-like feature crossing the dam crest and downstream dam face. Instead, during the period when most of the reservoir drains, the upstream, or proximal, part of the breach channel is steeply inclined and chute-like, while the downstream, or distal, part of the breach channel is gently inclined (Figure 2). The "breach crest"—the head of the breach channel—is defined by connecting the highest points on the dam that lie along streamlines passing through the breach channel [*Andrews*, 1998, p. 169; *Coleman et al*., 2002, p. 832]. The breach crest therefore coincides with a hydraulic control section that separates subcritical flow upstream (where the flow is contained between the walls of the experimental flume) from supercritical flow downstream (where the flow is contained within the self-formed breach channel). It is the size and shape of this arcuate hydraulic control section that determine discharge from the reservoir.

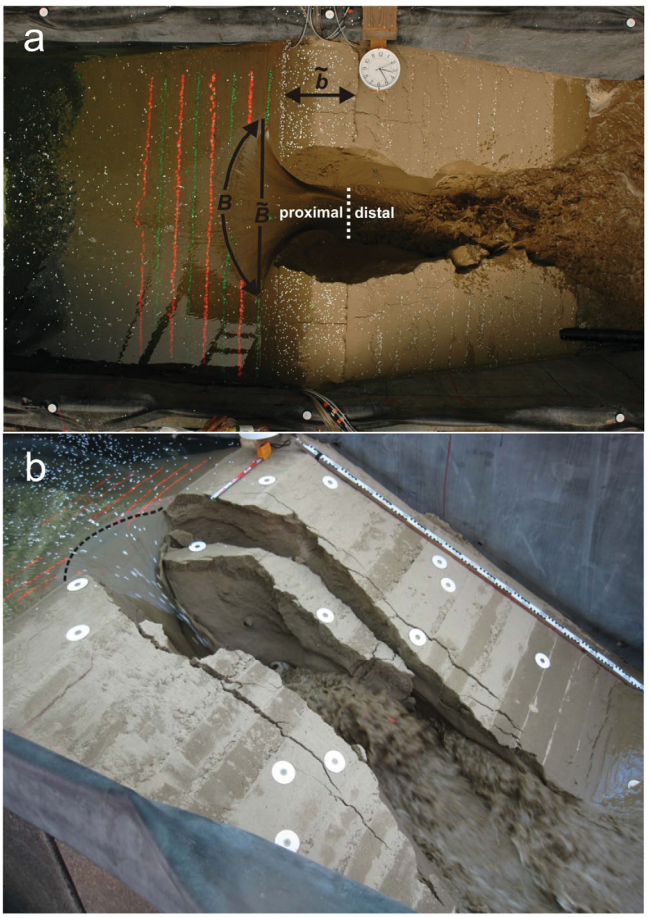


Figure 2. Two snapshots from our dam-breaching experiments showing that the breach geometry is not well described by Figure 1. (a) Overhead view showing that the channel head, or breach crest, where hydraulic control occurs, is arcuate, with length B. One possible definition of breach width is $\tilde{B}$, the straight-line distance between the ends of the breach crest. The breadth of the dam crest will be denoted $\tilde{b}$. (b) Oblique view showing that flow is supercritical and seemingly free of suspended sediment as it plunges over the breach crest (dashed line), but becomes highly turbid upon encountering the distal hydraulic jump. White disks are DVDs placed on the dam surface to serve as photogrammetric targets.

The earliest description of the formation of an arcuate hydraulic control feature during dam breaching seems to have been by *Gordon* [1981, 1990], who investigated the drainage of an intermittently closed lagoon near Sydney, Australia. The first documentation of breach-crest evolution in the laboratory appeared in two theses from the University of Auckland [*Jack*, 1996;

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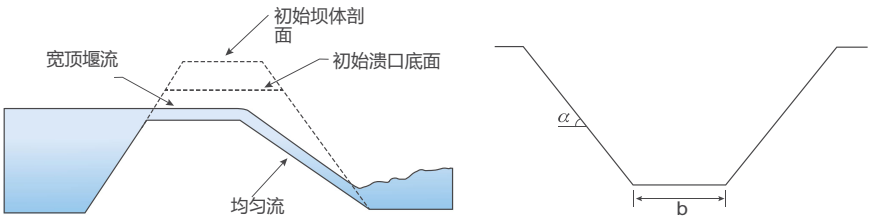


图1．参数化模型中常被不恰当地假定的溃口几何形态[经许可引自吴等，2009，国际水利环境工程学会]。水流沿一条沟槽流动，该沟槽首先穿越正在侵蚀的坝顶（此处假定存在水力控制），继而流经下游坝面，沟槽底部处发生水跃。沟槽的横断面形状（右图）通常取为梯形，底宽为b，边坡为a。

参数化溃坝模型的价值主要在于：相较于地貌动力学模型，其计算工作量与成本大幅降低，却仍能揭示计算所得洪水过程线如何受表征溃口演化各类参数的影响。当然，这仅在参数化模型正确表征物理系统时才有可能实现。然而——此处我们提前引出下文将更详细呈现的结果——当人们观察实际溃坝事件本身，而非仅其事后痕迹时，

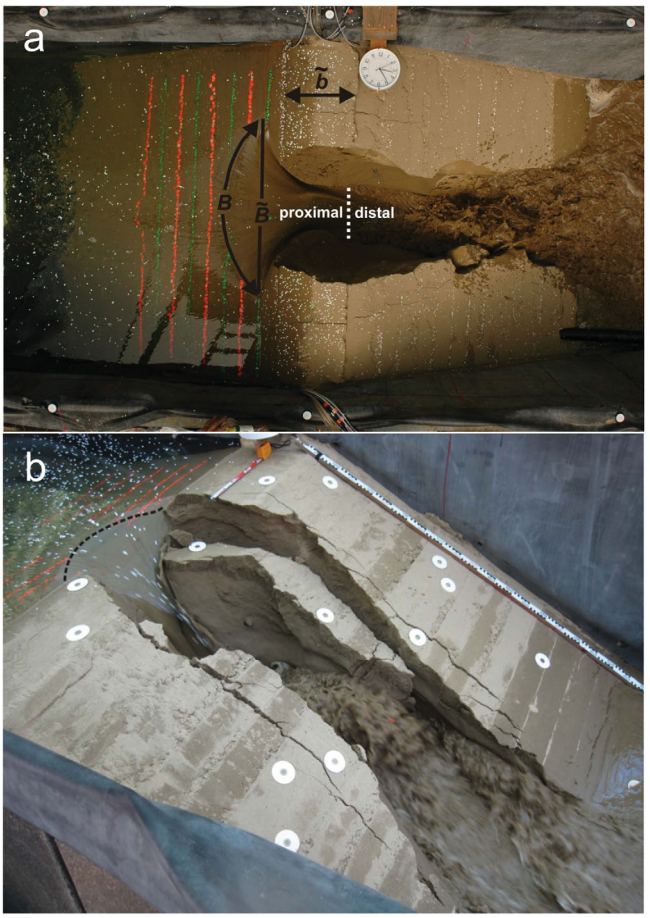


图2．我们开展的溃坝实验中两个瞬间状态快照，表明溃口几何形态无法由图1恰当描述。（a）俯视图显示，发生水力控制的通道顶端（即溃口顶部）呈弧形，长度为B。溃口宽度的一种可能定义是~B，即溃口顶部两端之间的直线距离。坝顶宽度记为~b。（b）斜视图显示，水流在越过溃口顶部（虚线所示）时呈超临界流态且貌似不含悬移质，但遭遇远端水跃后即变得高度浑浊。白色圆盘为置于坝面的DVD光盘，用作摄影测量靶标。

由此可明显看出，在坝顶及下游坝面存在沟槽状特征期间，实际排泄的水量极少。相反，在水库大部分水量排出的时段内，溃口通道的上游段（即近端段）坡度陡峭、呈急槽状，而下游段（即远端段）坡度则较为平缓（图2）。溃口顶部——即溃口通道的顶端——定义为沿穿过溃口通道的流线所经过的坝顶最高点的连线[安德鲁斯，1998，第169页；科尔曼等，2002，第832页]。因此，溃口顶部恰好对应一个水力控制断面，该断面将上游亚临界流（水流被实验水槽侧壁所约束）与下游超临界流（水流被自行形成的溃口通道所约束）分隔开来。正是这一弧形水力控制断面的尺寸与形状，决定了水库的泄流量。

关于溃坝过程中弧形水力控制特征形成的最早描述，似乎出自戈登[1981，1990]，他研究了澳大利亚悉尼附近一处间歇性封闭潟湖的排水过程。实验室中溃口顶部演化的首次记录见于奥克兰大学的两篇学位论文[杰克，1996；

Andrews, 1998]; these studies were subsequently summarized in peer-reviewed literature by *Coleman et al.* [2002]. Jack and Andrews utilized an experimental setup that permitted the abrupt shut-off of drainage from a reservoir (the level of which was held constant) and subsequent inspection of the morphology of the breached dam (constructed of either sand or fine gravel). By building a series of identical dams, breaching them, and allowing the experiments to proceed for progressively longer intervals Δt_i (i=1 to N), where $\Delta t_1 < \Delta t_2 < \Delta t_3, \ldots, \Delta t_{N-1} < \Delta t_N$, Jack and Andrews developed what amounted to proxy data sets that gave breach-crest shape as a function of time during an uninterrupted experiment. By also noting "shorelines" etched into the dam [*Andrews*, 1998, pp.161–162], the water-surface elevation above the breach crest was estimated and thus an approximate flow cross section could be determined for each Δt_i. In planform, the breach crest resembled a so-called minimum-energy-loss weir [*Chanson*, 2004b]; the flow cross section [*Andrews*, 1998, pp. 167–184] was long and slender, quite unlike the cross section of any conventional weir used for discharge measurement.

In the remainder of this paper, we report results of a set of large-scale dam-overtopping experiments that were designed to (i) directly measure time-dependent evolution of the breach-crest cross section, (ii), assess reproducibility of the flood hydrograph in near-replicate experiments, and (iii) investigate the effect of changing initial dam height and water level on the flood hydrograph. Determination of the breach crest morphology and the flow cross section at the breach crest was done using photogrammetric methods complemented by video records, including from an underwater camera. We show conclusively that the breach-crest cross section, which controls outflow from the reservoir, does not represent a trade-off between erosional and depositional processes, as in existing parametric breaching models, but is rather a purely erosional feature shaped predominantly by tractional erosion. We show that modest differences between flood hydrographs in nominally replicate experiments probably reflect subtle differences in dam hydrology from one experiment to another, and that the observed systematic variation in peak discharge with initial water level (or equivalently dam height) can be understood in terms of hydrological and mechanical processes that shape the breach channel.

Two thesis studies recently done at the University of Ottawa [*Orendorff*, 2009; *Al-Riffai*, 2014] explored themes that overlap with our work. Both studies used photogrammetric methods to estimate or measure breach crest shape and cross section. *Orendorff* [2009] explored the effect on the flood hydrograph of changing the shape of the initial breach in the dam crest; *Al-Riffai* [2014] demonstrated that close attention to construction methods led to reproducible flood hydrographs, and also conducted experiments with dams of different heights to explore whether Froude scaling applied to flood hydrographs. We will discuss some of Al-Riffai's results later.

We do not discuss here the mechanics of earthen-dam failure induced by groundwater seepage or piping. While there have undoubtedly been earthen dams that failed in this fashion [*Froehlich*, 1987], the likely importance of material heterogeneities in the process makes experimental repeatability a vexing problem. The reader interested in piping failure of earthen dams is referred to *Wahl and Lentz* [2011].

2. Experimental Setup and Materials

Thirteen experiments with homogeneous dams were conducted in the course of three campaigns (in 2010, 2012, and 2013) at the U.S. Geological Survey debris-flow flume in the H.J. Andrews Experimental Forest near Blue River, Oregon, USA [*Iverson et al.*, 2010]. The flume, which is a 2 m wide rectangular chute on a steep hillslope, debouches onto a nearly planar concrete surface with an average slope of 2.4° (Figure 3). We temporarily extended the flume side walls by erecting a series of 1.2 m high concrete panels braced against toppling with large concrete blocks. A 1.14 mm thick sheet of EPDM rubber was draped over the walls and bed to form a water-tight reservoir. Dams were built atop the EPDM sheet and bounded laterally by the concrete panels (Figure 3). The extended flume walls ended about 0.2 m beyond the dams' downstream toe, so the escaping flood flow was almost entirely unconstrained, unlike the situation in a typical laboratory flume. To reduce the likelihood of seepage failure, we installed a horizontal toe drain from wall to wall at $x \approx -1.1$ m (Figure 4). The drain consisted of 90 mm outside diameter ABS pipe that was perforated and covered with bronze filter cloth. Water that entered the drain was conveyed by another pipe to a discharge point several meters downstream.

Dam construction proceeded as follows: A set of steel brackets was bolted to the concrete walls at an angle of ~30° with the horizontal. Wooden tongue-and-groove boards were then slid into the brackets and the

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安德鲁斯，1998]；这些研究随后由科尔曼等人[2002]在同行评议文献中进行了综述。杰克与安德鲁斯采用了一种实验装置，使水库（其水位保持恒定）的排水能够突然中断，并对溃坝后的坝体形态（由砂或细砾石筑成）进行后续观测。通过建造一系列完全相同的坝体、逐一使其溃决，并让各次实验分别持续 progressively longer intervals Dti（i从1至N），其中Dt1 < Dt2 < Dt3;...; DtN−1 < DtN，杰克与安德鲁斯构建出若干组代理数据集，用以表征一次连续实验过程中溃口顶部形态随时间的变化规律。此外，他们还记录了刻印在坝体上的岸线痕迹[安德鲁斯，1998，第161–162页]，据此估算溃口顶部以上的水面高程，从而为每个Dti确定近似的过流断面。在平面形态上，溃口顶部形似所谓“最小能量损失堰”[尚松，2004b]；其过流断面[安德鲁斯，1998，第167–184页]狭长而纤细，与任何用于流量测量的传统量水堰断面均截然不同。

本文余下部分报告一系列大规模漫顶溃坝实验的结果，其设计目标为：（i）直接测量溃口顶部横断面随时间的动态演化；（ii）评估近似重复实验中洪水过程线的可重复性；（iii）探究初始坝高与库水位变化对洪水过程线的影响。溃口顶部形态及溃口处过流断面的确定，采用摄影测量法并辅以视频记录（含水下摄像）。我们明确证实：控制水库出流的溃口顶部横断面并非现有参数化溃口模型所假设的侵蚀与淤积过程权衡结果，而纯粹是牵引侵蚀主导形成的侵蚀性地貌。我们还表明：名义上重复实验中洪水过程线的微小差异，可能源于各次实验间坝体水文条件的细微差别；而观测到的峰值泄流量随初始库水位（或等效坝高）系统性变化现象，可借助塑造溃口通道的水文与力学过程予以解释

渥太华大学近期完成的两项学位论文研究（Orendorff，2009；Al-Riffai，2014）涉及与本工作重叠的主题。两项研究均采用摄影测量法估算或测定溃口顶部形态与横断面。Orendorff（2009）探讨了坝顶初始溃口形状变化对洪水过程线的影响；Al-Riffai（2014）证实，严格把控施工工艺可获得可重复的洪水过程线，并开展不同坝高的实验以检验弗劳德相似律是否适用于洪水过程线。后文将讨论Al-Riffai的部分结果

本文不讨论由地下水渗流或管涌引发的土坝失事机理。尽管确有土坝因此类原因失事（Froehlich，1987），但该过程中材料非均质性的重要影响使实验可重复性成为棘手难题。对土坝管涌失事感兴趣的读者可参阅Wahl与Lentz（2011）

2. 实验装置与材料

在美国地质调查局（USGS）位于俄勒冈州蓝河附近H.J.安德鲁斯实验林的泥石流槽中，分三个阶段（2010、2012及2013年）共开展了13组均质坝实验（Iverson等，2010）。该水槽为宽2 m的矩形陡坡溜槽，末端汇入近乎水平的混凝土面，平均坡度2.4°（图3）。我们临时加高水槽侧墙：竖立若干高1.2 m的混凝土板，并以大型混凝土块加固防倾覆。将一张厚1.14 mm的EPDM橡胶垫覆盖于侧墙及床面，形成密闭水库。坝体建于EPDM垫上，两侧由混凝土板限界（图3）。加高后的水槽侧墙末端延伸至坝体下游坝趾外约0.2 m，因而溃流几乎完全不受约束，不同于典型实验室水槽情形。为降低渗流破坏可能性，我们在x ≈ −1.1 m处沿墙至墙设置了一条水平坝趾排水管（图4）。该排水管为外径90 mm的ABS穿孔管，外包青铜滤布。进入排水管的水经另一根管道导流至下游数米处的出水口

坝体施工步骤如下：一组钢制支架以约30°倾角螺栓固定于混凝土侧墙；随后将木制企口板插入支架中，并……



Figure 3. A view from downstream of the experimental area. The dam, which is built upon an impermeable rubber sheet, is confined by walls 2 m apart and 1.2 m high, but unconfined downstream. The reservoir depth is maximum at the dam and tapers to zero upstream.

space between them was filled with clean, well-sorted, moist beach sand (D_{50}=0.21 mm) that was dumped from a front-end loader, distributed using rakes and shovels, sprinkled with water, and finally compacted using a vibratory compactor of the type commonly used for landscaping and small-scale construction. The sediment layers produced in this way were about 0.1 m thick with a smooth, flat, nearly horizontal surface. After taking sediment samples and placing sensors (discussed below), the process was repeated until the desired dam height (Table 1) had been reached, at which point we smoothed and leveled the dam crest. Finally, the framework was unbolted from the flume walls and removed. Dams built in this manner had a moderate degree of variability in water content and dry bulk density (see the tabulation of sediment properties in Table 2), as would any natural dam or indeed any constructed dam at prototype scale.

The downstream toe of the dam defines x=0 in our local Cartesian coordinate system (Figure 4). The origin of coordinates is the point where that toe touches the left-hand sidewall (as viewed looking upstream). The x and y axes are in the horizontal plane and the z axis is vertical. Note that all dams were built at the same location with a toe-to-toe distance L_b of 3.5 m, so that the lower the dam, the greater was the breadth $\tilde{b}$ of the dam crest and the ratio $\tilde{b}/L_b$ (Table 1, Figure 2a). Owing to the 2.4° slope of the basal surface on which dams were built, the height of the dam crest relative to the downstream toe (H_d) exceeded the height relative to the upstream toe (H_u) by 0.15 m.

Three still-frame cameras (Nikon D70 digital single-lens-reflex, fitted with a 35mm Nikkor lens set at f/8 and manually focused) were positioned ∼5 m overhead and along the centerline of the embankment (y=1 m), spaced ∼1.4 m apart to provide approximately 60% overlap of the fields of view of adjacent cameras, enough for stereo coverage of the entire dam. The cameras were wired to a common remote-control cable that activated all camera shutters simultaneously.

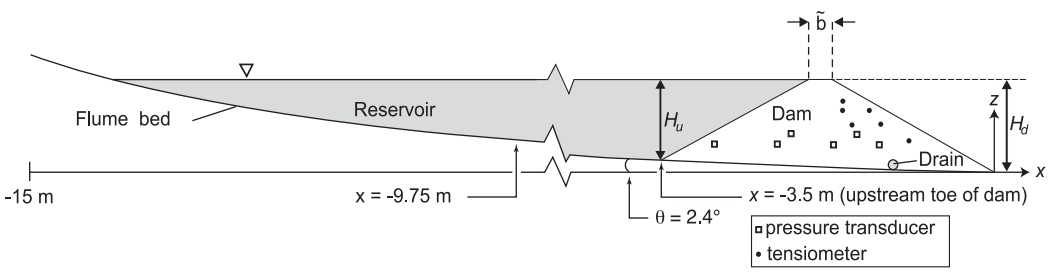


Figure 4. Schematic but properly scaled drawing of dam and reservoir. The dam is built on a gently sloping concrete pad. Note that H_u and H_d, the elevation of the dam crest relative to the upstream toe and downstream toe, respectively, are unequal: H_u=H_d−0.15 m. For the example shown here, H_d=0.98 m. The width $\tilde{b}$ of the dam crest varied depending upon H_d (see Table 1). Total reservoir length varied with dam height. Positions of sensors within the dam are appropriate to Experiment 10.

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图3. 实验区下游视角图。坝体建于不透水橡胶垫上，两侧由相距2 m、高1.2 m的侧墙约束，下游无约束。水库水深在坝前最大，向上游逐渐减小至零

其间空隙填筑洁净、级配良好、含水湿润的海滩砂（D50 = 0.21 mm），由前端装载机倾倒，再以耙子和铁锹摊铺、洒水润湿，并最终采用常用于园林绿化及小型工程的振动式压路机压实。由此形成的沉积层厚约0.1 m，表面平滑、平整且近乎水平。在采集沉积物样品并布设传感器（后文详述）后，重复上述工序直至达到目标坝高（表1），最后整平并修整坝顶。最终，将框架从水槽侧壁拆卸并移除。此类方法建造的坝体，其含水率与干容重存在中等程度变异性（参见表2中沉积物特性统计），这与天然坝体乃至原型尺度人工坝体的情形一致

下游坝趾定义为本地笛卡尔坐标系中的x = 0（图4）。坐标原点为该坝趾与左侧边墙（面向上游观察时）的接触点。x轴与y轴位于水平面内，z轴垂直向上。需注意：所有坝体均在同一位置建造，坝趾间距Lb恒为3.5 m，因此坝体越低，坝顶宽度~b及其与Lb之比~b/Lb越大（表1，图2a）。由于坝体所建基底表面倾角为2.4°，故坝顶相对于下游坝趾的高度（Hd）比其相对于上游坝趾的高度（Hu）大0.15 m

三台静态摄像机（尼康D70数码单反相机，配35 mm尼克尔镜头，光圈f/8，手动对焦）安装于坝体上方约5 m处，沿坝轴线（y = 1 m）布置，相邻相机间距约1.4 m，使相邻视场重叠率约60%，足以实现整个坝体的立体观测。各相机通过一根共用远程控制线缆同步触发快门

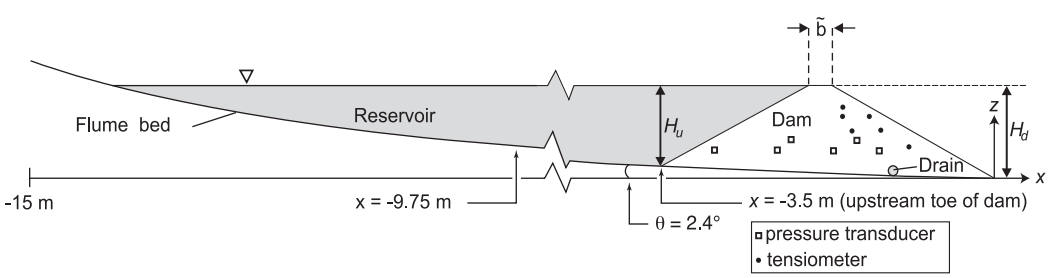


图4. 坝体与水库示意图（按实际比例绘制）。坝体建于缓倾混凝土基座上。注意：Hu和Hd分别为坝顶相对于上游坝趾和下游坝趾的高程，二者不等：Hu = Hd − 0.15 m。本图所示示例中，Hd = 0.98 m。坝顶宽度~b随Hd变化（见表1）。水库总长随坝高变化。坝体内传感器布设位置对应实验10

Table 1. Dam and Reservoir Characteristics, and Peak Discharge[a]

Experiment[b]	Date	H_d[c]	H_u[d]	Z_0[e]	$\tilde{b}$[f]	$\tilde{b}/L_b$[g]	A_0[h]	V_0[i]	Q_{peak}[j]
1	18 Aug 2010	1.00	0.85	0.98	0.36	0.10	23.7	12.74	0.126
2	20 Aug 2010	1.00	0.85	0.98	0.36	0.10	23.7	12.74	0.130
3	24 Aug 2010	1.00	0.85	0.98	0.36	0.10	23.7	12.74	0.137
4	26 Aug 2010	1.00	0.85	0.98	0.36	0.10	23.7	12.74	0.172
5	13 Jun 2012	1.02	0.87	0.99	0.33	0.09	23.8	12.95	0.148
6	14 Jun 2012	0.76	0.61	0.75	1.16	0.33	20.5	7.82	0.087
8	20 Jun 2012	0.58	0.43	0.58	1.75	0.50	18.26	4.60	0.041
9	21 Jun 2012	0.57	0.42	0.58	1.78	0.51	18.07	4.42	0.043
10	5 Jun 2013	1.00	0.85	0.98	0.36	0.10	23.7	12.74	0.134
11	6 Jun 2013	0.75	0.60	0.74	1.19	0.34	20.33	7.62	0.072
12	10 Jun 2013	0.87	0.72	0.87	0.74	0.21	22.22	10.33	0.138
13	12 Jun 2013	0.87	0.72	0.87	0.74	0.21	22.22	10.33	0.149
14	13 Jun 2013	0.66	0.51	0.67	1.47	0.42	19.3	6.08	0.054

[a]All dams were built on a surface sloping at 2.4°, with a toe-to-toe distance L_b of 3.5 m, and had faces sloped at an angle of 30° to the horizontal. The distance between the confining walls was 2.0 m.
[b]Experiment 7 was run with a layer of gravel and cobbles within the dam and is not discussed in this paper.
[c]H_d = initial dam height (m) relative to downstream toe.
[d]H_u = initial dam height (m) relative to upstream toe.
[e]Z_0 = maximum reservoir level (m) relative to downstream toe.
[f]$\tilde{b}$ = initial breadth of dam crest (m) in the mean flow direction.
[g]$\tilde{b}/L_b = \tilde{b}/3.5$ m, breadth of dam crest divided by toe-to-toe length of the dam.
[h]A_0 = initial surface area of reservoir (m^2).
[i]V_0 = initial volume of reservoir (m^3).
[j]Q_{peak} = peak discharge ($m^3\ s^{-1}$).

To obtain continuous video documentation of each experiment, we used at least five high-definition digital video cameras (nominal frame rate of 30/s) for simultaneous, overlapping recordings. Images from one of these cameras (mounted next to the central still-frame camera) were used for particle image velocimetry (PIV). Two video cameras placed in front of the flume provided detailed oblique views of the dam, while another was set up to provide an oblique view of the dam from upstream. Finally, one video camera was placed within a Plexiglas box and suspended in the reservoir from a mounting bracket placed across the flume. This underwater box was weighted so as to be neutrally buoyant and descend as the reservoir drained. This camera provided an underwater view of the upstream dam face and the developing breach crest.

Piezometers (Honeywell Microswitch model 6PCCFA3D, 0–0.1 MPa range) and tensiometers (UMS model T5) were installed during dam construction by gently excavating previously compacted sand, placing the sensors, and finally backfilling the cavities and compacting them by hand. In the 2010 campaign, we simply used a measuring tape and level to locate these sensors relative to lines of x=constant and z=constant drawn on the EPDM rubber sheet. We estimate placement error for the 2010 campaign to be about 0.05 m in both the horizontal and vertical. In 2012 and 2013, we again used tape and level for an approximate location, but in addition took stereopair photographs of the scene so that sensor locations could later be determined photogrammetrically, to within about 0.01 m in the horizontal and vertical. The piezometers were vented to the atmosphere and mounted within a short piece of PVC pipe covered with a piece of fine-mesh brass screen, an arrangement that permitted direct pressure communication between pore water and piezometer diaphragm while also protecting the diaphragm from clogging or mechanical damage. Experience with these piezometers [*Iverson et al.*, 2010] has shown that instrumental drift is a potential problem, so we typically recalibrated them for each experiment. Reservoir level was measured using a pair of submersible transducers (Pressure Systems Inc. model KPSI 730) placed at the bottom of stilling wells.

All electronic sensor outputs were amplified near their sources, transmitted via shielded cables to a sheltered station near the base of the flume, and sampled digitally using a personal computer equipped with a data-acquisition card. Sensor voltages were sampled at a rate of at least 10 Hz.

Starting with Experiment 4, we placed lines of brightly painted pebble gravel on the upstream face of the dam, each line being approximately at some x=constant. As will be described in more detail below, as the reservoir drained, the breach crest retreated upstream, and pebbles were undermined as the sand on which they rested began to move (supporting information Movies S1 and S2). (Note that all Movies cited are found in the supporting information. Complete video summaries of the experiments [*Logan and Iverson*, 2007,

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表1. 坝体与水库特征及峰值泄流量 a 实验 b 日期

	H_d[c]	H_u[d]	Z_0[e]	$\tilde{b}$[f]	$\tilde{b}/L_b$[g]	A_0[h]	V_0[i]	Q_{peak}[j]
2010年8月18日								
2010年8月20日								
2010年8月24日								
2010年8月26日								
2012年6月13日								
2012年6月14日								
2012年6月20日								
2012年6月21日								
2013年6月5日								
2013年6月6日								
2013年6月10日								
2013年6月12日								
2013年6月13日								

a所有坝体均建于坡度为2.4°的坡面上，坝趾间距Lb为3.5 m，上下游坝坡与水平面夹角均为30°。两侧约束墙间距为2.0 m

b 实验7在坝体内铺设了一层砾石与卵石，本文未予讨论
c Hd：相对于下游坝趾的初始坝高（m）
d Hu：相对于上游坝趾的初始坝高（m）
e Z0：相对于下游坝趾的最高库水位（m）；f ~b：沿平均水流方向的坝顶初始宽度（m）
g ~b=Lb：坝顶宽度与坝趾间长度之比，~b=3.5 m
h A0：水库初始水面面积（m^2）
i V0：水库初始库容（m^3）
j Qpeak：峰值泄流量（m^3/s）

为全程连续记录每次实验过程，我们至少使用五台高清数字摄像机（标称帧率为每秒30帧）同步、交叉覆盖拍摄。其中一台摄像机（紧邻中央静态图像相机安装）所摄图像用于粒子图像测速（PIV）分析。两台置于水槽前方的摄像机提供坝体的详细斜向视图，另一台则从上游方向提供坝体斜向视图。最后，一台摄像机被置于

一个有机玻璃箱体内，并通过横跨水槽的支架悬吊于水库中。该水下箱体配重使其呈中性浮力，可随水库水位下降而同步下沉。该摄像机提供了坝体上游坡面及发育中溃口顶部的水下视角。

孔隙水压力计（Honeywell Microswitch型号6PCCFA3D，量程0–0.1 MPa）和张力计（UMS型号T5）在筑坝过程中安装：先小心挖开已压实的砂层，放入传感器，再回填空腔并手工压实。2010年实验中，我们仅使用卷尺和水平仪，依据铺设在EPDM橡胶板上的x = 常数与z = 常数参考线定位传感器。据此估算，2010年实验的传感器定位误差在水平与垂直方向均约为0.05米。2012年与2013年实验中，我们仍先用卷尺与水平仪进行粗略定位，另增加拍摄场景立体像对照片，以便后期通过摄影测量法精确确定传感器位置，其水平与垂直方向定位误差均控制在约0.01米以内。孔隙水压力计通大气，安装于一段短PVC管内，管口覆以细密黄铜网，该结构既保证孔隙水与压力计膜片直接传压，又可防止膜片堵塞或机械损伤。根据以往使用经验［Iverson等，2010］，仪器漂移是潜在问题，故我们通常在每次实验前重新标定压力计。水库水位采用一对浸没式变送器（Pressure Systems Inc.型号KPSI 730）测定，安装于消能井底部。

所有电子传感器输出信号均在靠近信号源处放大，经屏蔽电缆传输至水槽底部附近的遮蔽式工作站，并由配备数据采集卡的个人计算机进行数字化采样。传感器电压信号采样频率不低于10 Hz。

自实验4起，我们在坝体上游坡面上按近似x = 常数的位置布置多排颜色鲜明的卵石砾石。如后文详述，随着水库泄水，溃口顶部向上游后退，卵石因所依附的砂层开始运动而逐渐失去支撑（参见支持信息影片S1和S2）。［注：文中提及的所有影片均收录于支持信息中。全部实验的完整视频摘要［Logan和Iverson，2007，

Table 2. Sediment Properties[a]

Property	Value
D_{10}, D_{50}, D_{90}	0.17, 0.21, 0.26 (mm)
ρ_d, dry bulk density[b]	1514 ± 56 kg m^{-3} ($N = 100$)
ω, water content[b] (mass water/mass solids)	0.070 ± 0.024 ($N = 100$)
n, porosity[b]	0.43 ± 0.02 (N=100)
ϕ, static friction angle of dry sediment[c]	33.5 ± 1.1 deg (N=65)
w_s, grain settling speed in water[d]	0.024 m s^{-1}
K_{sat}, saturated hydraulic conductivity[e]	4.5×10^{-5} m s^{-1} at ρ_d=1600 kg m^{-3} 4.5×10^{-5} m s^{-1} at ρ_d=1625 kg m^{-3} 1.2×10^{-5} m s^{-1} at ρ_d=1660 kg m^{-3}

[a]The beach sand, which was supplied by Eugene Sand and Gravel of Eugene, Oregon, was sourced from a dune near Florence, Oregon. Tabulated values of measurements are given in supporting information Tables S1–S4. *N* denotes total number of measurements.
[b]For sand as placed and compacted during dam construction.
[c]Determined with a tilt table.
[d]Using equation (9) of *Cheng* [1997] with kinematic viscosity corresponding to a temperature of 15°C.
[e]Determined with a compaction permeameter; tabulated values are averages of two measurements. Note ρ_d values are greater than for the built dams.

revised 2013] are viewable at http://pubs.usgs.gov/of/2007/1315/.) Thus initiation of pebble movement, which could be seen in the overhead and underwater video, served as a reasonably accurate proxy for migration of the eroding breach crest along the upstream dam face.

3. Experimental Procedure

After construction of a dam, water was pumped into the reservoir. Depending upon pump deployment and dam height, filling time ranged from 6 to 54 min. During this period of time, we began logging data and recording video. Synchronization of sensor and video data was achieved by using audible and/or visual signals recorded by both systems.

The initial breach was a V-shaped notch across the flat dam crest, about 25 mm deep and 35 mm across, cut manually with a long piece of aluminum angle stock. In Experiments 1–3, out of a sense of caution about avoiding seepage instability, the notch was cut within 3–4 min of filling the reservoir. In subsequent experiments, we allowed infiltration to proceed while monitoring the appearance of the dam's downstream toe, signals from the piezometers, and outflow from the toe drain; we did not cut the notch until the piezometers nearest the drain had responded and outflow had risen to a roughly constant level (about 3 to 8 L/min).

As water escaped from the reservoir through the breach during overtopping, paper confetti (typical size 3–4 mm) was scattered on the water slightly upstream of the breach to help visualize the surface velocity field. In some experiments, especially those with relatively low, broad-crested dams, confetti blockages formed during the earliest stages of breaching as the stream lost water by infiltration, and were gently cleared using the angle stock. A small recharge to the reservoir (never more than 15 L/min, or $\sim 2.5 \times 10^{-4}$ m^3/s) was supplied in some experiments to offset infiltration losses. This recharge was trivial compared to the typical peak discharge Q_{peak} during breaching of $\sim$0.1 m^3/s, so our experiments may be regarded as using a fixed-volume reservoir. In comparison, laboratory studies of levee breaching are customarily configured with a constant-water-level reservoir of effectively infinite volume [e.g., *Pickert et al.*, 2011].

Six experiments utilized nearly identical $\sim$1 m high dams; we also ran pairs of replicate experiments at three lesser dam heights (Table 1). Although there have been many prior laboratory investigations of dam or levee breaching, few have included either near-replicate experiments or systematic variation of dam height. We will discuss some of those previous experiments later when we assess repeatability of results in our own experiments.

4. Data Analysis

Recorded sensor voltages were processed by averaging over a 2 s moving window, downsampling at a 2 s interval, and then converting to physical units using the sensor calibrations. For piezometers and tensiometers, gauge pressure p was converted to pressure head $\psi = p/\rho_w g$, where ρ_w is the density of water and g is the acceleration due to gravity. Total head h equals ψ plus sensor elevation relative to $z = 0$.

A time stamp was overlaid on each video recording, consistent with the previously mentioned synchronization of these recordings with the logged sensor data. Timing of selected still-camera images was then accomplished by comparing those images with time-stamped video from the overhead video

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表2．沉积物特性a 性质

	数值
D10；D50；D90	0.17、0.21、0.26（毫米）
ρd：干容重	1514±56 kg·m³（N = 100）
ω：含水率（水质量/固体质量）	0.070±0.024（N = 100）
n：孔隙率	0.436±0.02（N = 100）
φ：干沉积物静摩擦角	33.5°±1.1°（N = 65）
ws：颗粒在水中的沉降速度	0.024 m·s¹
Ksat：饱和水力传导度	4.5×10⁵ m·s¹（ρd = 1600 kg·m³）；4.5×10⁵ m·s¹（ρd = 1625 kg·m³）；1.2×10⁵ m·s¹（ρd = 1660 kg·m³）

a 本研究所用海滩砂由俄勒冈州尤金市Eugene砂石公司提供，采自俄勒冈州弗洛伦斯附近沙丘。各项测量值汇总于支持信息表S1–S4。N表示测量总次数。

b 指筑坝施工过程中砂料铺设并压实后的状态。
c 采用倾角台测定。
d 采用Cheng［1997］公式（9）计算，运动粘度对应温度为15°C。

e 采用压实式渗透仪测定；表列数值为两次测量的平均值。注意：ρd值高于建成坝体实测值。

修订版2013]可在http://pubs.usgs.gov/of/2007/1315/查阅。因此，卵石起动现象（可在俯视及水下视频中清晰观察到）可作为溃口侵蚀顶部沿坝体上游坡面迁移的一个合理且较准确的代用指标。

3．实验步骤

坝体建成后，用水泵向水库注水。注水时间随水泵布置方式及坝高而异，介于6至54分钟之间。在此期间，我们即开始记录传感器数据及视频。传感器数据与视频数据的同步通过两种系统共同记录的声学和/或光学信号实现。

初始溃口是在平坦坝顶上人工切割出的V形缺口，深约25毫米、宽约35毫米，使用一根较长的铝制角钢完成。在实验1–3中，出于对渗透失稳风险的谨慎考虑，缺口在水库充水完毕后3–4分钟内即被切开。在后续实验中，我们允许渗流持续发生，同时监测坝体下游坡脚处的渗出现象、孔隙水压力计读数以及坡脚排水口的出流情况；仅当靠近排水口的孔隙水压力计已产生响应、且出流上升并趋于稳定（约3–8升/分钟）后，才开始切割缺口。

当水库水体经溃口漫顶溢出时，纸制彩纸（典型尺寸3–4毫米）被撒布于水面以辅助可视化表面流速场。为辅助可视化水面流速场，在溃口稍上游的水面上撒布了彩纸（典型尺寸约4毫米）。在部分实验中——尤其是坝体相对低矮且顶部较宽的实验中——溃决初期因渗流导致水流减小，彩纸易堆积堵塞，此时采用铝制角钢轻柔清除。部分实验中还向水库补充少量水量（始终不超过15升/分钟，即约2.5×10^{4} m³/s），以补偿渗流损失。该补水量远小于溃决过程中的典型峰值流量Qpeak（约0.1 m³/s），因此本实验可视为固定库容水库。相比之下，堤防溃决的实验室研究通常采用恒定水位水库，其有效库容可视为无限大［例如Pickert等，2011］。

共进行了六组实验，所用坝体高度均接近1米；此外，还在三种较低坝高下各开展了两组重复实验（见表1）。尽管此前已有大量关于坝体或堤防溃决的室内实验研究，但其中极少包含近似重复实验，亦鲜有系统性地改变坝高的研究。后文在评估本实验结果的可重复性时，将对部分先前研究作进一步讨论。

4．数据分析

记录的传感器电压信号经2秒滑动窗口平均处理，并以2秒间隔降采样，再依据传感器标定系数换算为物理量单位。对于孔隙水压力计和张力计，将测得的表压p转换为压力水头w = p / (ρw g)，其中ρw为水的密度，g为重力加速度。总水头h等于w加上传感器相对于z = 0基准面的高程。

每个视频记录均叠加了时间戳，该时间戳与前述传感器数据记录的同步方式一致。选定的静止相机图像的时间定位，则通过将其与俯视视频的时间戳进行比对完成。

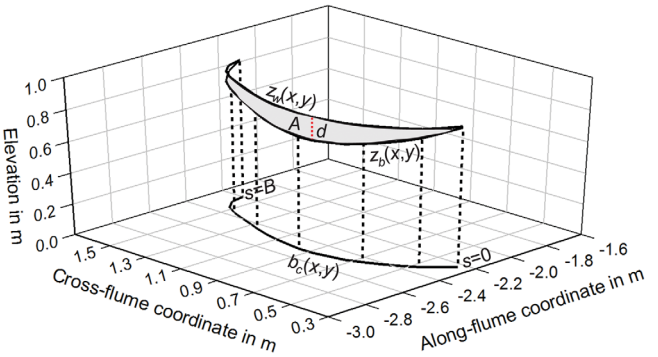


Figure 5. Illustration of method for determining the breach crest cross section. The curve labeled $b_c(x,y)$ represents the interpolated (x,y) coordinates of the breach crest at some time t_i. The interpolation derives from data collected on the initiation of pebble motions. Overlaying this curve on the water surface and dam surface gives the curves $z_w(x,y;t_i)$ and $z_b(x,y;t_i)$, respectively. The red-dotted line indicates local flow depth $d=z_w-z_b$, and the shaded region is the total flow cross sectional area, A. This drawing depicts the actual flow cross section at an instant in time during Experiment 5 when $Q \approx 0.11\ \mathrm{m}^3/\mathrm{s}$.

camera. By matching the locations of moving objects in the still photos with those in the video, the timing of the still images could be resolved to within 1 video frame, or 0.033 s.

We used particle-image velocimetry (PIV) to determine the water-surface velocity field in several of our experiments. PIV does not involve tracking individual tracer particles, but rather relies on cross-correlation between sequential images to determine average particle displacement over finite areas (cells). Knowing the time between images and the function that maps image space onto physical space, one can finally calculate local velocities. For our dam-breach experiments, the PIV analysis was done using the overhead video and tracer confetti on the water surface. Details of the process are described in supporting information Text S1.

Photogrammetric analysis of selected stereo pairs of still-camera images was done using 3-DM Analyst Research Suite 2.3 software (Adam Technology, Perth, Australia). Using camera calibration values derived by the software, relative-only 3-D points were automatically calculated, resulting in a sparse cloud of ~1000 points defining the surface of the water and unsubmerged parts of the dam. For Experiments 1–4, data were oriented and properly scaled with respect to our local coordinate system by introducing, during the photogrammetric analysis, a set of control-point coordinates calculated from dam geometry. An improved approach used in later experiments was to scale and orient the relative-position data by using surveyed control points that had been placed within the fields of view of all three cameras. After fitting the sparse, relative-only point clouds to our local coordinate system, we generated digital orthophotos and densely populated triangular irregular network (TIN) digital terrain models (~60,000 3-D points) of the water and dam surfaces. DTMs from adjacent pairs of cameras were merged to produce a single image that included the entire dam and the adjoining reservoir surface. Some TIN data were also interpolated onto a regular grid and used to make contour maps, for which purpose we used Tecplot software. Photogrammetrically derived elevations could be compared with values of reservoir level calculated from the water-level-sensor data. The two methods generally agreed within 0.01 m, which we take as the error in the photogrammetrically derived elevations.

The location of the breach crest as a function of time, required for later discussion of the evolution of the breach crest, was determined by using the video record of pebble motions. (This method worked well for some experiments but not all: it failed when the colored pebbles became buried by shifting sand on the submerged dam face during prolonged initial periods of low discharge.) We began by comparing the time-stamped video with a rectified overhead image of the uneroded dam with pebbles present. Let $t=T_j$ be the time of initial motion of a pebble with coordinates $(x=X_j, y=Y_j)$, and suppose that we make the same determination for a large number of pebbles. Mathematically, this set of numbers $T_j(X_j, Y_j)$ may be thought of as sampled from a surface $T(x,y)$, which represents the time of initial pebble motion everywhere. A contour on this surface for, say, time t_i—found by interpolating from the $T_j(X_j, Y_j)$—is a curve in the horizontal plane whose (x,y) coordinates are those of the breach crest $b_c(x,y;t_i)$. We next (Figure 5) overlaid $b_c(x,y;t_i)$ on the dam surface $z_d(x,y)$ to get $z_b(x,y;t_i)$, the elevation of the breach crest at t_i. Similarly overlaying $b_c(x,y;t_i)$ on the water surface $z_s(x,y)$ gave $z_w(x,y;t_i)$, the elevation of the water surface directly above the breach crest at time t_i. The local flow depth along the breach crest was then $d(x,y;t_i)=z_w(x,y;t_i)-z_b(x,y;t_i)$. To calculate the total flow cross sectional area at the breach crest, $A(t_i)$, we expressed d as a function of arc length s along the breach crest, with $s=0$ at one end and $s=B_i(t)$ at the other:

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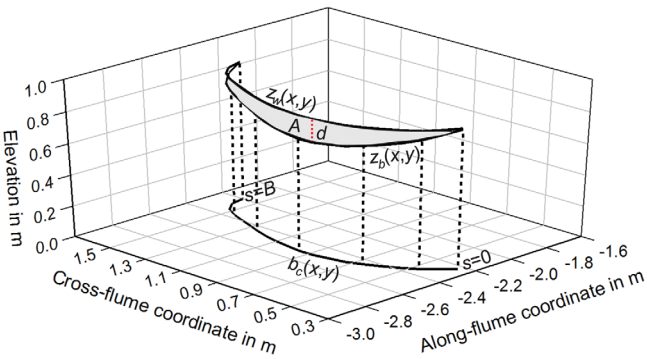


图5．确定溃口顶部横断面的方法示意图。标注为bc(x, y)的曲线表示某一时刻ti下溃口顶部插值得到的x、y坐标。该插值基于卵石起动过程观测数据获得。将此曲线叠加于水面线及坝体表面线上，分别得到zw(x, y, ti)和zb(x, y, ti)两条曲线。红色虚线表示局部水流深度d = zw − zb，阴影区域则代表总过流横断面面积A。本图描绘的是实验5进行过程中某一瞬时的实际过流横断面，此时流量Q ≈ 0.11 m³/s。

相机。通过比对静止照片与视频中运动物体的位置，可将静止图像的时间精度确定至1帧视频，即0.033秒。

我们采用粒子图像测速法（PIV）测定若干次实验中水面的速度场。PIV不依赖于追踪单个示踪颗粒，而是通过连续图像间的互相关分析，确定有限区域（即计算单元）内颗粒的平均位移。已知图像采集的时间间隔以及图像空间到物理空间的映射关系，即可最终计算出局部流速。在本研究的坝体溃决实验中，PIV分析工作已完成

利用俯视视频和撒在水面上的示踪彩纸。该过程的详细信息见支持信息文本S1。

采用3-DM Analyst Research Suite 2.3软件（Adam Technology公司，澳大利亚珀斯）对选定的静态相机立体像对进行摄影测量分析。利用软件导出的相机标定参数，自动计算仅具相对关系的三维点，生成约1000个点的稀疏点云，以定义水面及坝体未淹没部分的表面。对于实验1–4，通过在摄影测量分析中引入一组基于坝体几何尺寸计算得出的控制点坐标，将数据定向并按本地坐标系正确缩放。后续实验采用了更优方法：利用布设于全部三台相机视场内的实测控制点，对相对位置数据进行缩放与定向。在将稀疏的、仅具相对关系的点云拟合至本地坐标系后，我们生成了数字正射影像及密集填充的三角不规则网络（TIN）数字地形模型（约60,000个三维点），分别表征水面与坝体表面。相邻相机对生成的TIN数据经融合后，形成一幅涵盖整座坝体及毗邻水库水面的单一影像。部分TIN数据还被内插至规则网格，并用于制作等高线图（使用Tecplot软件完成）。摄影测量所得高程值可与水位传感器数据推算的库水位值进行比对，两种方法结果通常相差在0.01 m以内，此即摄影测量高程值的误差。

为便于后续讨论溃口顶线演化，需确定溃口顶线随时间变化的位置。我们利用卵石运动的视频记录确定该位置。（此方法在部分实验中效果良好，但并非全部适用：当彩色卵石在低泄流初期长时间淹没于坝面移动的沙层之下时，该方法失效。）首先，我们将带时间戳的视频与一幅校正过的俯视图像（未侵蚀坝体表面布设卵石）进行比对。设t = Tj为坐标x = Xj、y = Yj的卵石初始运动时刻，并假设对大量卵石均作出相同判断。数学上，该组数值Tj(Xj, Yj)可视为从曲面T(x, y)采样所得，该曲面代表各处卵石的初始运动时刻。对该曲面取t = ti的等值线，即得曲线bc(x, y, ti)。接着（图5），将bc(x, y, ti)叠加于水面zw(x, y, ti)之上，得到溃口顶线在ti时刻的高程zb(x, y, ti)。同样，将bc(x, y, ti)叠加于坝体表面zb(x, y, ti)之上，则得溃口顶线在ti时刻的高程zb(x, y, ti)。为计算溃口顶线处总过流断面面积A(ti)，我们将d表达为沿溃口顶线弧长s的函数，其中一端s = 0，另一端s = Bi(t)：

$$A(t_i) = \int_0^{B(t_i)} d(s; t_i)\, ds \tag{1}$$

We estimate the error in z_w and z_b to be about 15–20 mm in both horizontal and vertical, reflecting both photogrammetric error and uncertainty in picking the initiation of pebble motion. The error in d is the root-mean-square of the errors in z_w and z_b, or roughly 20–28 mm. The fractional errors in d and A are therefore substantial—about 15% at peak discharge and more at lower discharge.

Note that our tabulated values of B (supporting information Data set S1) are actually values of the arc length of $b_c(x, y; t_i)$, which is commonly about 3% less than the arc length of $z_b(x, y; t_i)$. We have chosen to do this to allow comparison to data of *Coleman et al.* [2002] and *Al-Riffai* [2014], whose measures of breach-crest length reflect projection onto a horizontal plane.

5. Results

5.1. Morphological Evolution of the Breach Channel

Breach development may be conveniently divided into stages, much as done by previous workers for both noncohesive sediment dams [*Visser*, 1994; *Visser et al.*, 1995; *Coleman et al.*, 2002] and cohesive sediment dams [*Hunt et al.*, 2005; *Hanson et al.*, 2005; *Morris and Hassan*, 2009]. Our description emphasizes certain points that are particularly important to the flood hydrograph. The referenced Movies are key to understanding the processes involved in breach evolution.

Stage I: The initial notch across the top of the dam is slowly incised as a stream of water begins to escape from the reservoir. As the dam is not fully saturated, the stream loses water by infiltration and its sediment load is deposited on the downstream dam face, building an alluvial fan that advances toward the toe of the dam (Figure 6a, supporting information Movie S3). The stream eventually incises the alluvial fan, forming a narrow channel, with nearly vertical banks, that extends headward to the breach crest. The channel gradient continuously adjusts as incision occurs and sediment is deposited on the downstream dam face, thereby extending the alluvial fan. The period of alluvial-fan growth and slow channel-slope adjustment lasts tens of minutes in experiments with relatively low, broad-crested dams, but no more than 1–2 min in experiments with narrow-crested dams. Invariably the channel develops zones of alternating steep and gentle gradient (Figure 6b), probably equivalent to "erosional cyclic steps" [*Parker and Izumi*, 2000], which form spontaneously in regions of supercritical flow. This stage ends when the incipient channel reaches the toe of the downstream dam face (which, in geomorphic terms, is the local base level), causing a step to form in the stream profile at the toe of the dam and establishing a pathway for sediment to be carried away. Meanwhile, at the channel head, an arcuate breach crest develops, with flow there initially subcritical, but becoming critical as discharge increases and the reach immediately downstream of the breach crest steepens to produce supercritical flow.

Stage II: The zones of alternating gradient along the dam face evolve into a stepped longitudinal profile. The steps migrate upstream, increase in amplitude, and commonly coalesce (supporting information Movie S4). The end stage of the step-migration process is a steep reach (for convenience termed a "headcut" hereinafter) that eventually merges with the breach crest, the location of hydraulic control (Figure 6c). The slope of the headcut is, as we will show later, at or near the static angle of friction of the sediment. Step migration is commonly disrupted by slumping of blocks of sediment from the banks of the breach channel, which are nearly vertical owing to the apparent cohesion produced by capillary forces [*Pickert et al.*, 2011]. Local highs in the water-surface elevation appear to be associated with hydraulic jumps, which we interpret as being paired with a series of plunge pools. (This pattern of migrating steep reaches and pools was in fact first described by *Gordon* [1981, 1990] in regards to breaching of coastal sand barriers).

Stage III: The headcut is eroded along its entire length by water pouring over the breach crest. In this way, the breach crest is undercut and migrates upstream and downward, becoming progressively longer, and runaway behavior commences, with discharge increasing rapidly as a result. The breach channel, meanwhile, is widened as blocks of sediment slump from the channel banks, with small blocks carried away by the stream, while the largest ones lodge in the stream channel and are progressively eroded. Slumping commonly alternates from one channel bank to the other, and the size of slump blocks tend to increase as

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$$A(ti) = \int_{0}^{(B(ti))} d(s, ti)ds \quad (1)$$

我们估算zw和zb的误差在水平与垂直方向均为约15–20 mm，这反映了摄影测量误差及判定卵石起动时刻的不确定性。d的误差为zw和zb误差的均方根，约为20–28 mm。因此，d与A的相对误差相当显著——峰值泄流时约15%，低泄流时更高。

需注意，我们汇总的B值（支持信息数据集 S1）实际为bc(x, y, ti)的弧长，通常比zb(x, y, ti)的弧长小约3%。我们选择如此处理，以便与Coleman等人［2002］及Al-Riffai［2014］的数据进行对比——他们对溃口顶线长度的测量均反映为水平面投影。

5. 结果

5.1 溃口河道形态演化

溃口发育过程可便捷地划分为若干阶段，此前学者对无黏性泥沙坝［Visser，1994；Visser 等人，1995；Coleman 等人，2002］及黏性泥沙坝［Hunt 等人，2005；Hanson 等人，2005；Morris 和 Hassan，2009］均采用过类似划分方式。我们的描述强调了若干对洪水过程线尤为关键的要点。所引述的影片对于理解溃口演化过程至关重要。

第一阶段：当一股水流开始从水库逸出时，坝顶初始切口缓慢下切。由于坝体未完全饱和，水流经下渗损失水量，其挟带泥沙则沉积于下游坝面，形成向坝趾推进的冲积扇（图6a，支持信息影片 S3）。水流最终切入冲积扇，形成一条狭窄、近垂直岸坡的河道，并向上游延伸至溃口顶线。随着下切进行及泥沙在下游坝面沉积（从而延伸冲积扇），河道坡度持续调整。在相对低矮、宽顶坝体的实验中，冲积扇生长及河道坡度缓慢调整阶段持续数十分钟；而在窄顶坝体实验中，该阶段不超过1–2分钟。河道 invariably 发育出交替出现的陡坡与缓坡区段（图6b），其性质可能等同于侵蚀性周期阶坎［Parker 和 Izumi，2000］，这类阶坎在超临界流区域可自发形成。当初始河道抵达下游坝面坝趾（地貌学上即当地基准面）时，该阶段结束：坝趾处河道纵剖面形成阶坎，为泥沙运移开辟了通道。与此同时，在河道源头处发育出弧形溃口顶线，此处初始为亚临界流，但随泄流量增大及溃口顶线下游河段变陡而转为超临界流。

第二阶段：坝面上交替变化坡度的区域演化为阶梯状纵向剖面。这些阶坎向上游迁移、振幅增大，并常发生合并（支持信息影片 S4）。阶坎迁移过程的最终阶段是一段陡坡段（为方便起见，后文称其为“溯源侵蚀陡坎”），该陡坡段最终与溃口顶线合并，而溃口顶线即为水力控制点（图6c）。如后文所示，陡坎坡度等于或接近泥沙的静摩擦角。阶坎迁移常因溃口河道两岸泥沙块体滑塌而中断——这些岸坡近乎垂直，系毛细力产生的表观黏聚力所致［Pickert 等人，2011］。水面局部抬升现象似与水跃相关，我们将其解释为一系列跌水潭的对应表现。（事实上，这种迁移式陡坡段与跌水潭的组合模式，最早由 Gordon［1981，1990］在描述海岸沙坝溃决时提出。）

第三阶段：水流越过溃口顶线，沿整个陡坎长度进行侵蚀。由此，溃口顶线被掏蚀，并向上游及向下迁移，长度持续增加，进而引发失控行为——泄流量随之急剧增大。与此同时，溃口河道因两岸泥沙块体滑塌而拓宽：较小块体被水流带走，最大块体则滞留于河道内并持续遭受侵蚀。滑塌常交替发生于河道两岸，且滑塌块体尺寸随……而增大。

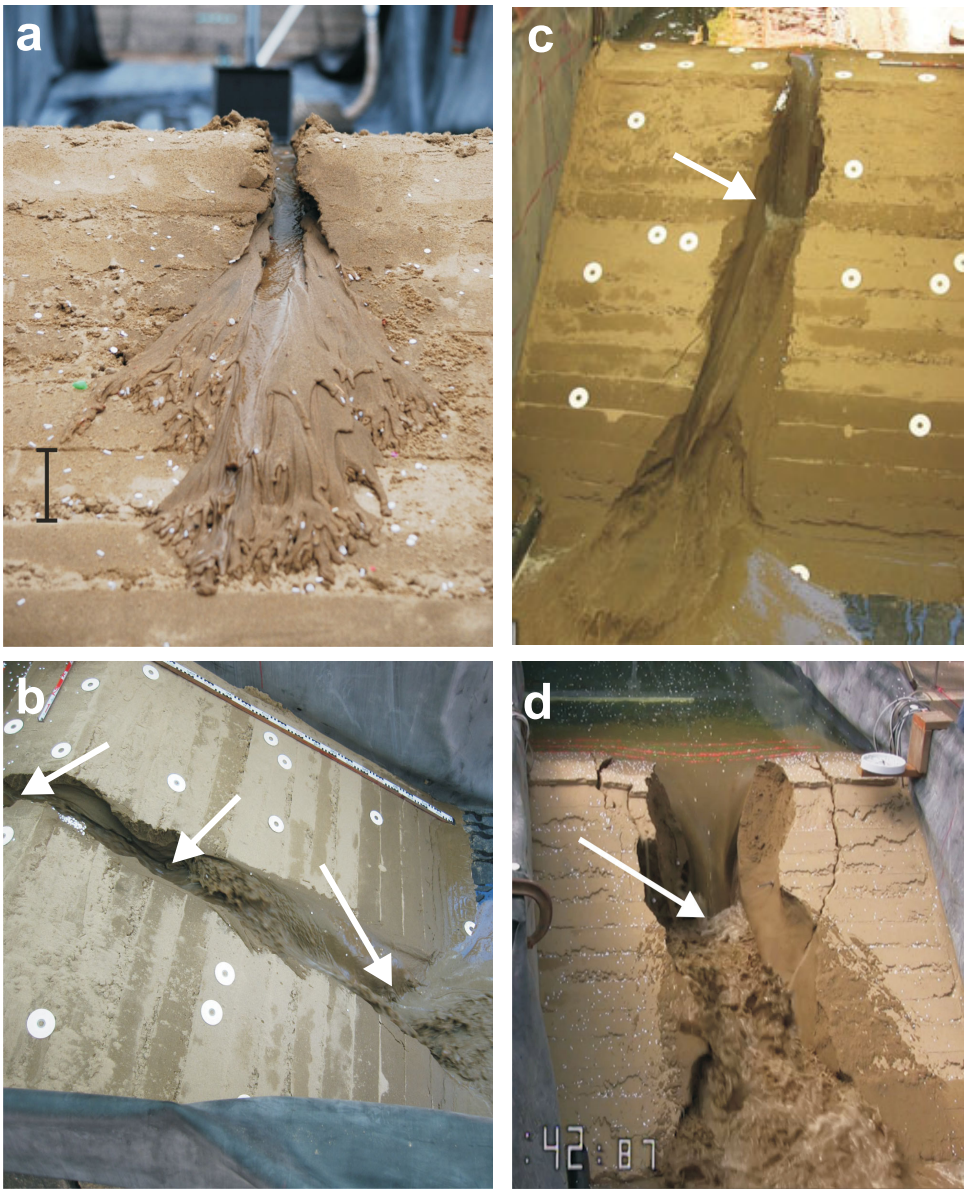


Figure 6. Progressive stages of dam breaching in an experiment with a 0.98 m high dam. (a), (c), and (d) are views from downstream. (a) Earliest stage of dam breaching. An alluvial fan is constructed as the incipient streamflow infiltrates the downstream dam face and deposits its sediment load. Scale bar indicates a distance of about 0.13 m along the sloping dam face. (b) Development of erosional cyclic steps along the dam face. The incipient channel has by this point in time incised its alluvial fan and extended to the toe of the dam. White arrows indicate transition from steeply dipping supercritical flow to gently dipping subcritical flow. Total length of the dam face is about 2 m. (c) Development of a headcut by migration and coalescence of cyclic steps. White arrow indicates base of headcut and likely position of a plunge pool. Scale as for Figure 6b. (d) Crack-bounded sediment blocks that are about to slump into the breach channel. White arrow indicates base of headcut and likely position of plunge pool. Note that flow downstream of the putative plunge pool is highly turbid. Scale as for Figure 6b.

time progresses (supporting information Movie S5). On rare occasions, bank failures extend far enough upstream to cause an abrupt lengthening of the breach crest (Figure 6d). Supporting information Movie S6 (underwater) shows that such perturbations are quickly smoothed out by tractional erosion, which dominates growth of the breach crest.

In most of our experiments, erosion proceeds to the bed of the flume before the reservoir has fully drained. In this case, the breach may continue to widen as residual water drains. Sometimes erosion does not proceed completely to the bed because sediment accumulated upstream of the drain pipe tends to create a tailwater effect.

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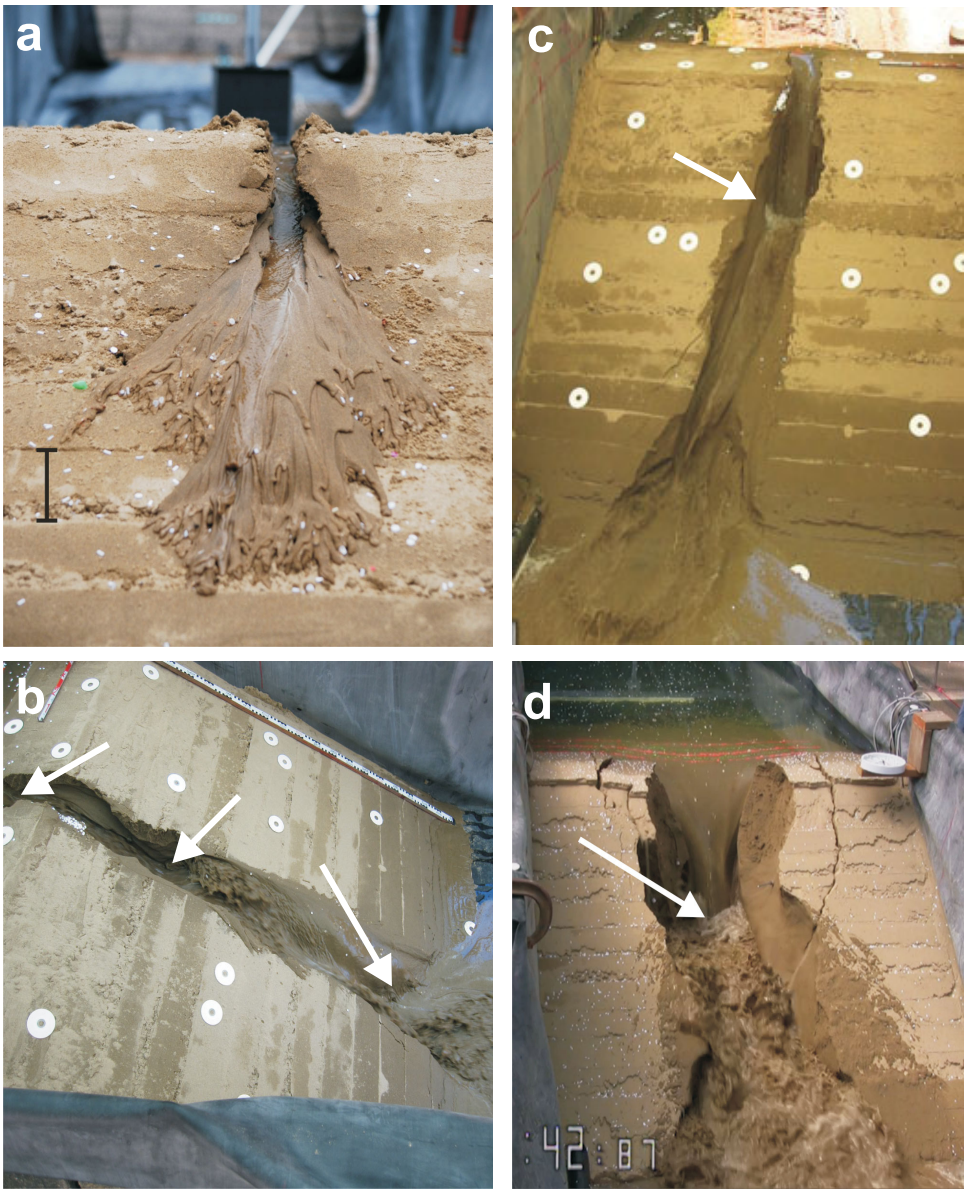


图6. 一座高0.98 m大坝溃决实验的渐进阶段。（a）、（c）和（d）为下游视角。（a）溃决最早阶段：初始水流渗入下游坝面并沉积其挟带的泥沙，形成冲积扇。比例尺表示沿倾斜坝面约0.13 m的距离。（b）坝面上侵蚀性周期阶坎的发展。此时，初始河道已切入冲积扇，并延伸至坝趾。白色箭头指示由陡倾超临界流转变为缓倾亚临界流的过渡位置。坝面总长约2 m。（c）通过周期阶坎的迁移与合并发育形成的溯源侵蚀陡坎。白色箭头指示陡坎底部及潜在的跌水潭位置。比例尺同图6b。（d）即将滑塌入溃口河道的、受裂缝约束的泥沙块体。白色箭头指示陡坎底部及潜在的跌水潭位置。注意，假想跌水潭下游的水流高度浑浊。比例尺同图6b。

随着时间推移（支持信息影片 S5）。极少数情况下，岸坡失稳向上游延伸较远，导致溃口顶线突然延长（图6d）。支持信息影片 S6（水下拍摄）显示，此类扰动会迅速被牵引侵蚀所抹平，而牵引侵蚀主导着溃口顶线的发展。

在我们的大部分实验中，侵蚀在水库完全排空之前已发展至水槽床面。此时，残余水流排出过程中，溃口可能继续展宽。有时，由于排水管上游堆积的泥沙倾向于形成尾水效应，侵蚀无法完全抵达床面。

The lower and broader the original dam crest, the greater is the opportunity for infiltration of stream water there, and the more prolonged is stage I. The same stages of breach development were seen for all dams except, to a certain extent, for Experiment 13, which unfolded differently owing to the fact that part of the downstream dam face began to liquefy and creep. As the incipient channel grew and encountered the liquefied toe of the dam, the stream flow spread out instead of incising. Over time, an alluvial fan was constructed (supporting information Movie S7). Eventually a step retreated to the upstream dam face and triggered enhanced erosion at the breach crest; the increasing discharge led to incision of the alluvial fan. Another example (from the IMPACT study [*Morris et al*., 2007]) of a breaching experiment with a liquefied toe may be seen in supporting information Movie S8: in that case, the entire downstream half of a narrow-crested dam progressively slumped until one such failure breached the dam crest. The escaping water then quickly incised the liquefied sediment. The key point here is that liquefaction of the toe does not necessarily lead to rapid breaching.

It should be clear from the Movies that breach evolution is an intrinsically three-dimensional phenomenon. The morphodynamics of a two-dimensional experiment, in which water spills over the entire embankment simultaneously, are distinctly different [see, e.g., *Chinnarasri et al*., 2003; *Schmocker and Hager*, 2009]: stage I is highly compressed, and stage III is, by definition, absent. Two-dimensional experiments can be highly instructive for understanding erosion mechanics, but *by definition* cannot capture the process of breach widening and development of the arcuate breach crest.

5.2. Long Profile Evolution

Three examples of the evolving water surface profile through the breach channel, approximately following the centerline of the flow, are shown in Figure 7: Experiment 5 with an initial dam-crest breadth $\tilde{b}$ of 0.33 m, Experiment 6 with $\tilde{b}=1.16$ m, and Experiment 9 with $\tilde{b}=1.78$ m. In all three experiments, there was a period during which the breach channel was more or less pinned at both the downstream dam toe and the upstream edge of the dam crest (marked by an X), but underwent a gradual adjustment owing to net erosion. This period, the duration of which increased as dam height decreased, ended once a step retreated to intersect the upstream dam face (i.e., the start of stage III). In Experiment 9 ($\tilde{b}/L_b=0.50$), the resulting headcut rather quickly became muted (supporting information Movie S9), whereas in Experiments 5 and 6 ($\tilde{b}/L_b=0.1$ and 0.33, respectively), the headcut was an enduring feature. It is interesting to compare these results with those of *Das* [1997], the only other investigator, as far as we know, to have reported morphological development in 3-D overtopping experiments with broad-crested cohesionless dams ($\tilde{b}/L_b \geq 0.2$). *Das* (1997, p. 89) presented two figures showing profiles of the sediment surface along the flow centerline for one dam with $\tilde{b}/L_b=0.46$, another with $\tilde{b}/L_b=0.67$. In both cases, the breach channel was initially pinned at the upstream edge of the dam crest; in neither case is a migrating headcut indicated.

In Experiments 5 and 6, the water-surface slope below the breach crest during stage III tended to a roughly constant value. In a 2-D experiment [e.g., *Chinnarasri et al*., 2003; *Schmocker and Hager*, 2009], the water surface would be nearly parallel to the bed surface. In a 3-D breaching experiment like ours, however, the strong lateral flow convergence is analogous to the situation in a convergent chute spillway, with the water surface slope exceeding the bed slope at least near the entrance to the convergent reach [*Reinauer and Hager*, 1998]. This detail notwithstanding, the fact that the water-surface slope remains nearly constant as erosion proceeds suggests that the bed surface must be maintaining a nearly constant slope, θ_r, close to the static friction angle of the sand. A similar result has been found in other experiments with fine sand and a fixed-volume reservoir [*Spinewine et al*., 2004; *van Emelen et al*., 2011, 2014].

To understand why $\theta_r \approx \phi$, consider a parcel of water spilling over the breach crest. The parcel accelerates and thus the shear stress τ it exerts on its bed (the headcut) rises monotonically with distance from the breach crest. If one supposes that the local erosion rate is everywhere a monotonically increasing function of τ [e.g., *Fujisawa et al*., 2008], then the bed would continuously steepen. This is in fact what happens during overtopping of cohesive sediment dams [e.g., *Hanson et al*., 2005; *Fujisawa et al*., 2011]. For noncohesive dam material such as beach sand, however, the slope along the flow centerline, where the sediment is undoubtedly saturated and unaffected by capillary forces, is generally limited to ϕ, with any steepening beyond this point counteracted by avalanching of bed sediment that restores the slope to a value close to ϕ.

In the case of a reservoir either maintained at a constant water level [*Andrews*, 1998] or recharged at a large, constant rate [*Schmocker*, 2011], the upstream dam face and the dam crest are eroded so quickly that the

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原始坝顶越低、越宽，河道水流在此处入渗的机会越大，第一阶段持续时间也越长。除实验13在某种程度上有所不同外，所有坝体均表现出相同的溃口发展阶段；实验13因部分下游坝面开始液化并蠕动，其演化过程有所区别。随着初始渠道扩大并遭遇液化坝趾，河道水流并未下切，而是横向扩散。随时间推移，形成了冲积扇（支持信息影片S7）。最终，一道台阶后退至上游坝面，触发溃口顶部加剧侵蚀；流量增加继而导致冲积扇被下切。另一例溃口实验（来自IMPACT研究[Morris等, 2007]）中坝趾液化的情形见于支持信息影片S8：在该案例中，一座窄顶坝体的整个下游半部逐步发生滑塌，直至某次此类失稳事件导致坝顶溃决。随后逸出的水流迅速下切液化沉积物。此处的关键在于：坝趾液化未必导致快速溃决。

从影片中可清晰看出，溃口演化本质上是一种三维现象。二维实验的形态动力学特征截然不同——此时水流同时漫过整个坝体[参见，例如Chinnarasri等, 2003；Schmocker与Hager, 2009]：第一阶段高度压缩，而第三阶段按定义根本不存在。二维实验对理解侵蚀机理极具启发性，但按定义无法再现溃口展宽及弧形溃口顶部形成的过程。

5.2. 纵剖面演化

图7展示了溃口通道内随时间演化的水面纵剖面三例，大致沿水流中心线绘制：实验5（初始坝顶宽度~b为0.33米）、实验6（~b = 1.16米）及实验9（~b = 1.78米）。在全部三项实验中，均存在一段时期：溃口通道在下游坝趾与坝顶上游边缘（以X标记）两处基本固定，但因净侵蚀作用而经历渐进式调整。该时段持续时间随坝高降低而延长；一旦台阶后退并与上游坝面相交（即第三阶段开始），该时段即告结束。在实验9中（~b/L_b = 0.50），所形成的溯源侵蚀沟头较快趋于平缓（支持信息影片S9），而在实验5与实验6中（~b/L_b分别为0.1与0.33），该溯源侵蚀沟头则成为持久特征。将这些结果与Das[1997]的研究成果作对比颇为有趣——据我们所知，他是唯一另一位报道了宽顶无黏性坝（~b/L_b ≥ 0.2）三维漫顶实验地貌发育过程的研究者。Das（1997，第89页）给出了两张图，分别展示了一座~b/L_b = 0.46与另一座~b/L_b = 0.67的坝体沿水流中心线的沉积表面剖面。两种情况下，溃口通道初始均固定于坝顶上游边缘；且均未显示有迁移的溯源侵蚀沟头。

在实验5与实验6中，第三阶段溃口顶部以下的水面坡度趋于大致恒定值。在二维实验中[例如Chinnarasri等, 2003；Schmocker与Hager, 2009]，水面将几乎平行于床面。但在我们这类三维溃决实验中，强烈的侧向汇流则类似于收敛式溢洪道的情形，至少在汇流段入口附近，水面坡度大于床面坡度[Reinauer与Hager, 1998]。尽管存在这一细节差异，水面坡度在侵蚀过程中保持近乎恒定的事实表明，床面坡度h_r必然亦维持近乎恒定，且接近砂土的静摩擦角。其他采用细砂与定容水库开展的实验亦得出相似结论[Spinewine等, 2004；van Emelen等, 2011, 2014]。

为理解为何h_r ≈ φ，可考虑一股越过溃口顶部的水流。该水流团加速运动，因而其对河床（即溯源侵蚀沟头）施加的剪应力τ随距溃口顶部距离单调增大。若假设局部侵蚀速率处处为τ的单调增函数[例如Fujisawa等, 2008]，则河床坡度将持续变陡。事实上，黏性土坝漫顶时便发生此类情况[例如Hanson等, 2005；Fujisawa等, 2011]。但对于无黏性坝体材料（如海滩砂），在水流中心线处（此处沉积物无疑处于饱和状态且不受毛细力影响），坡度一般限于φ；一旦超过该值，床面沉积物即发生崩塌，使坡度恢复至接近φ的水平。

对于水位维持恒定[Andrews, 1998]或以较大恒定速率补给的水库[Schmocker, 2011]，上游坝面与坝顶侵蚀极快，以致

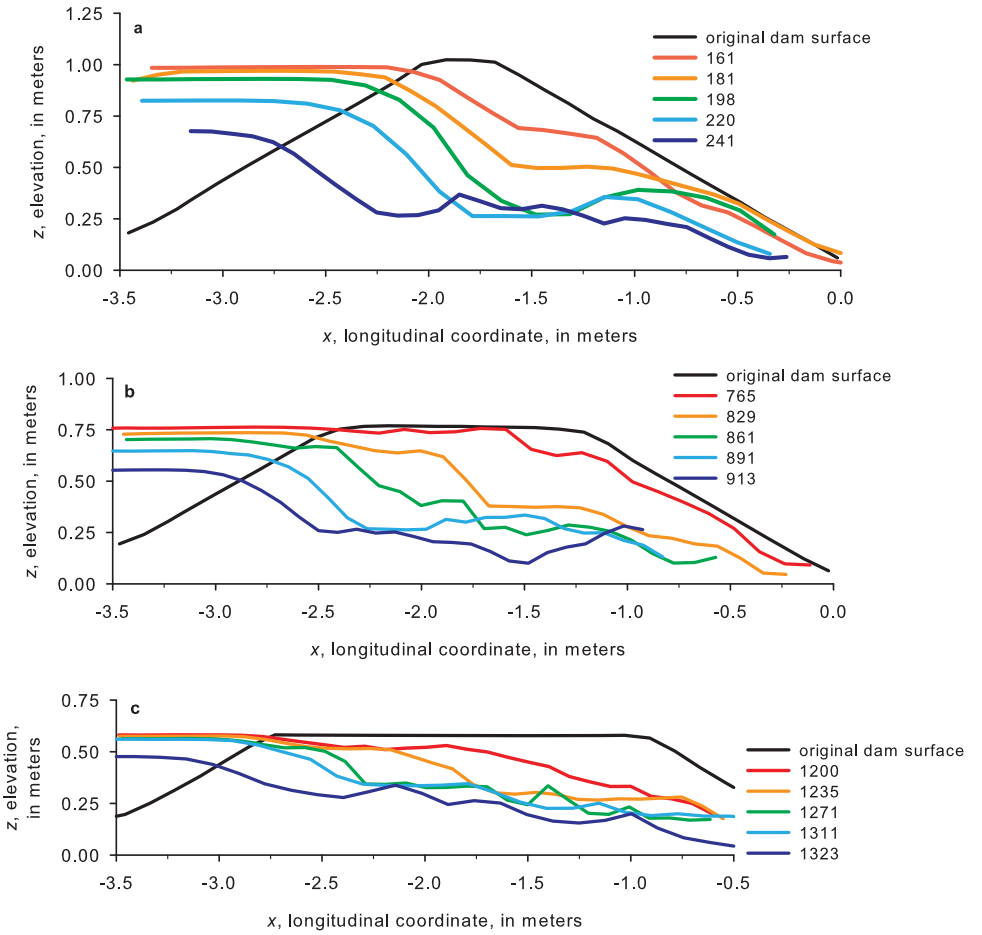


Figure 7. Profiles of the water surface, approximately following the centerline of the flow in (a) Experiment 5, (b) Experiment 6, and (c) Experiment 9. No vertical exaggeration. Colored curves represent photogrammetrically derived water-surface elevations and are labeled by the elapsed time in seconds since the notch was cut in the dam crest. At early time, before a headcut has migrated to the breach crest, the water surface elevation is more or less pinned at the rear of the dam crest (marked by an X) and the downstream toe. In Experiments 5 and 6, the headcut formed by retreat of a step to the breach crest maintained a steep angle, probably limited by the static angle of friction of the bed. In Experiment 9, in contrast, the headcut fairly quickly became muted.

slope of the downstream face never reaches ϕ. Over time the centerline profile takes on the shape of a low, broad sediment berm. Our Experiment 9 showed a similar style of morphological evolution.

5.3. Inferences About Sediment Transport

Flow through the proximal breach channel was supercritical and nearly always appeared to be free of suspended sediment, but the flow that emerged from the hydraulic jump into the distal breach channel was highly turbid (Figure 2). The only time that the flow through the proximal breach channel became turbid was when the flow encountered and eroded a slump block (supporting information Movie S5). Published figures in other studies of sand-dam breaching likewise show that the flow in the proximal breach channel is clear [*Andrews*, 1998 (Photos 5.7 and 5.8); *Hassan et al.*, 2004 (Figure 2); *Spinewine et al.*, 2004 (Figure 4); *Orendorff*, 2009 (Figure 5.39); *Cao et al.*, 2011, (Figure 5); *van Emelen et al.*, 2014 (Figure 11)]. It is tempting to conclude that what one sees in our experiments is the top of a bedload layer moving along the eroding dam face, as in the overtopping experiments of *Capart and Young* [2002] and *Berta and Bianco* [2010]. The situation is considerably more complicated, however: The threshold for suspension, typically expressed in terms of the ratio of u_*, the shear velocity, to w_s, the particle settling speed [*Chanson*, 2004a, pp. 183–84] is easily exceeded, yet the suspended sediment concentration is far from equilibrium. The bedload flux is also likely to be out of equilibrium. (These claims are backed by arguments presented in the supporting information.) This sediment-flux disequilibrium has consequences for scaling up laboratory experiments to field scale, discussion of which must be deferred to a later paper.

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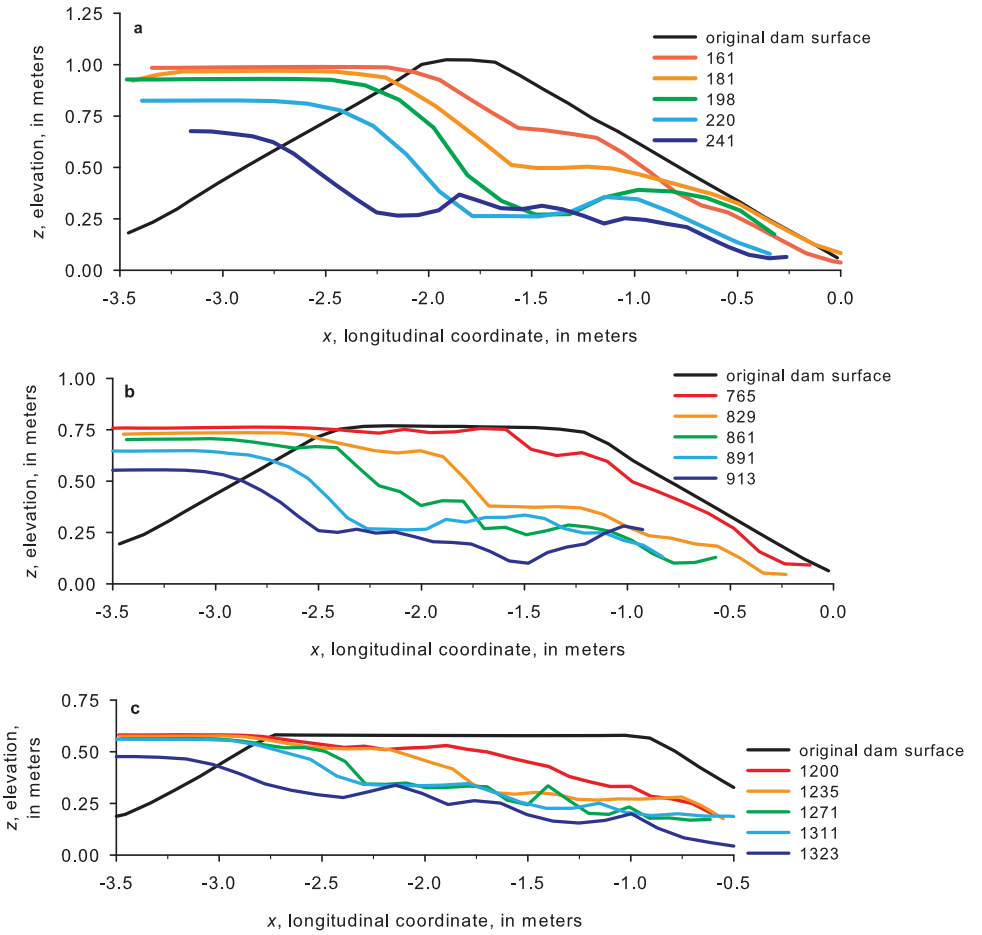


图7. 水面轮廓线，大致沿水流中心线绘制：(a) 实验5；(b) 实验6；(c) 实验9。未作垂直方向放大。彩色曲线代表通过摄影测量法反演得到的水面高程，并以大坝顶部缺口切开后经过的秒数标注。在早期阶段（即溯源侵蚀尚未抵达溃口顶部之前），水面高程大致固定于坝顶后缘（以X标记）及下游坝趾处。在实验5与实验6中，由台阶后退至溃口顶部所形成的溯源侵蚀沟头维持着陡峭倾角，此角度可能受床面静摩擦角限制。相比之下，在实验9中，该溯源侵蚀沟头则迅速趋于平缓。

下游坝面坡度从未达到φ。随着时间推移，中心线剖面逐渐呈现为低而宽的沉积土埂形态。我们的实验9也展现了类似的地貌演化模式。

5.3. 关于泥沙输运的推断

近溃口段渠道内的水流为急流，且几乎始终未见悬移质，但水流经水跃后进入远溃口段渠道时则高度浑浊（图2）。近溃口段渠道内水流唯一一次变浑浊的情况，是当水流遭遇并冲蚀一个滑塌块体时（支持信息影片S5）。其他关于砂质坝体溃决的研究文献中所发表的图件同样表明，近溃口段渠道内的水流是清澈的[Andrews, 1998（照片5.7与5.8）；Hassan 等, 2004（图2）；Spinewine 等, 2004（图4）；Orendorff, 2009（图5.39）；Cao 等, 2011（图5）；van Emelen 等, 2014（图11）]。人们很容易推断，我们在实验中所观察到的现象，正是沿侵蚀坝面移动的推移质层顶部，这与Capart和Young[2002]及Berta和Bianco[2010]开展的漫顶实验类似。然而实际情况却复杂得多：通常以剪切流速u*与颗粒沉降速度w_s之比表征的起悬阈值[Chanson, 2004a, 第183–184页]极易被超越，但悬移质浓度却远未达到平衡状态；推移质输沙率亦很可能处于非平衡状态。（这些论断均有支持信息中所列论证支撑。）这种泥沙输运率的非平衡状态，对实验室尺度实验向野外尺度的放大具有重要影响，相关讨论须留待后续论文展开。

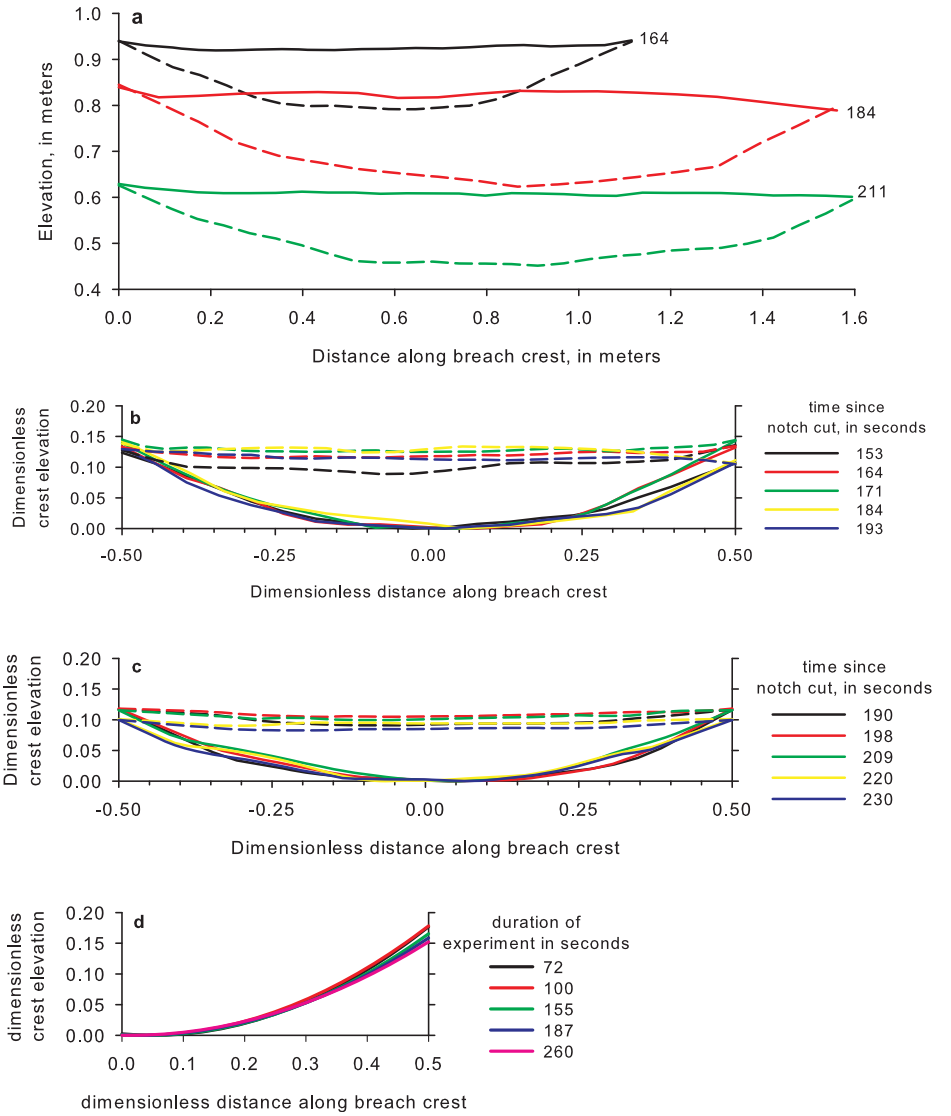


Figure 8. Vertical flow cross sections at the breach crest: (a) At three different moments during Experiment 4. Solid curves indicate water surface, dashed curves indicate breach crest. Black: 164 s after notch was cut. Red: 184 s after notch was cut. Green: 211 s after notch was cut. (b) Normalized vertical cross sections for Experiment 4. (c) Normalized vertical cross sections for Experiment 5. (d) Normalized vertical cross sections for one of *Andrews'* [1998] experiments, based on Andrews' tabulated polynomial fits to the shape rather than raw data.

5.4. Hydraulic Characteristics of the Breach Crest

Supporting information Movies S1 and S5 show that the vertical cross section at the breach crest is roughly parabolic in shape, and that the slope of the "banks" of the breach crest is considerably less than ϕ. However, the inherent foreshortening in these underwater views (because different points along the crest are at different distances from the camera) means that the Movies can only be used qualitatively. To get a quantitative picture of the breach shape and its temporal evolution, we use results from the photogrammetric and pebble-motion analyses described above. Examples of the evolving flow cross section at the breach crest are shown in Figure 8a. The shape of the cross section appears to change little with time. We can assess this idea more quantitatively by comparing dimensionless graphs of breach crest cross section. We define dimensionless distance $\tilde{s}$ along the breach crest as $\tilde{s}=s/B-1/2$, which ranges in value from $-1/2$ to 1/2. Appropriate dimensionless measures of breach-crest elevation and water-surface elevation are, respectively, $\left(z_b-z_{b,\min}\right)/B$ and $\left(z_s-z_{b,\min}\right)/B$, where $z_{b,\min}$ is the elevation of the lowest point along the breach crest. Plots of the normalized breach crest cross section for a series of snapshots in time are shown in

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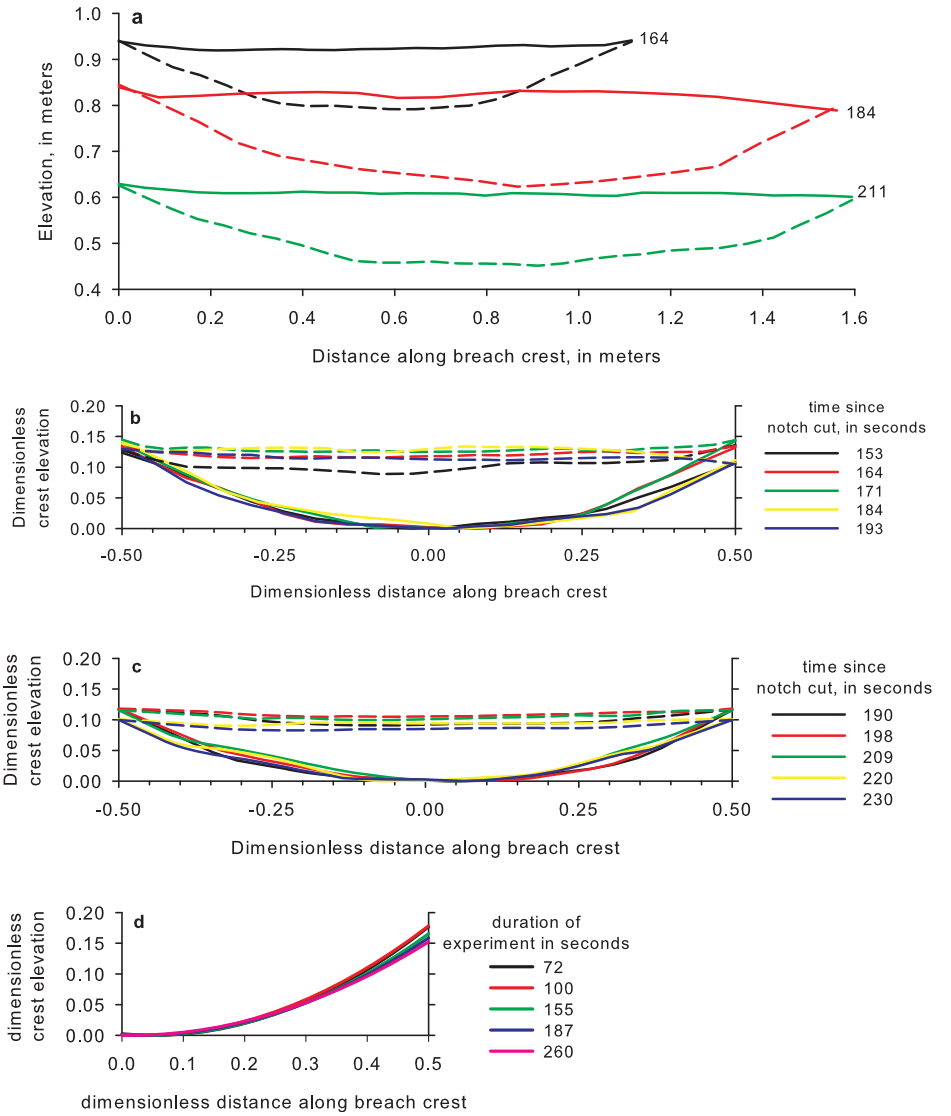


图8. 溃口顶部处的垂直水流过水断面：(a) 实验4进行过程中三个不同时刻的断面。实线表示水面，虚线表示溃口顶部。黑色：切开缺口后164秒；红色：切开缺口后184秒；绿色：切开缺口后211秒。(b) 实验4的归一化垂直断面。(c) 实验5的归一化垂直断面。(d) 基于Andrews[1998]实验中对形状拟合所得多项式系数（而非原始数据）绘制的归一化垂直断面。

5.4. 溃口顶部的水力特性

支持信息中的影片S1和S5显示，溃口顶部处的垂直过水断面大致呈抛物线形，且溃口顶部两岸边坡的坡度明显小于φ。然而，这些水下影像固有的透视缩短效应（因溃口顶部不同位置距相机距离各异）意味着这些影片仅能作定性分析。为定量刻画溃口形态及其随时间的演变，我们采用了前述摄影测量法与卵石运动分析所得结果。图8a展示了溃口顶部处随时间演化的水流过水断面示例。该断面形状似乎随时间变化甚微。我们可通过对比无量纲化溃口顶部过水断面图来更定量地验证这一观点。我们定义溃口顶部上的无量纲距离~s为~s = s / B^(1/2)，其取值范围为−1/2至1/2。溃口顶部高程与水面高程相应合适的无量纲化量分别为(z_b − z_{b,min}) / B 和 (z_s − z_{b,min}) / B，其中z_{b,min}为溃口顶部最低点的高程。一系列不同时刻快照所对应的归一化溃口顶部过水断面图见于

Figures 8b–8d for two of our experiments as well as one of *Andrews*' [1998] experiments. We see that the shape of the breach crest and the flow cross section are nearly invariant in time. The cross section is neither trapezoidal nor similar to the shape of any conventional weir used in engineering practice.

The $\langle\,\rangle$ brackets will be used to denote an average over the breach crest, that is,

$$\langle F\rangle=(1/B)\int_0^B F\,ds \tag{2}$$

for an arbitrary function F. The cross-sectional area A occupied by the flow at the breach crest may then be expressed as $A=\langle d\rangle B$. It is also useful to define a fictitious area $\hat{A}$ that represents the cross section that would exist in the absence of drawdown near the breach, namely,

$$\hat{A}=\int_{-\delta s_1}^{B+\delta s_2}\hat{E}(s)\,ds \tag{3}$$

where $\hat{E}=z_{ff}-z_b$ and δs_1, δs_2 are small corrections that represent slight changes in the "shoreline" position. The fictitious depth $\hat{E}$ is nearly equal to E, the total hydraulic head, or specific energy, of the reservoir relative to the breach crest, with an average value $\langle\hat{E}\rangle$ and maximum value $\hat{E}_{\max}$. The slight difference between $\hat{E}$ and E relates to the fact that flow lines at the breach crest are curved [*Castro-Orgaz and Chanson*, 2009]. As a practical matter, we may neglect the correction to $\hat{A}$ associated with δs_1 and δs_2, as it is small compared to the uncertainty in $\hat{A}$ associated with measurement error, as well as the difference between E and $\hat{E}$, and simply write $\hat{A}\approx\langle E\rangle B$. From data tabulated in the SI, we find

$$A/B^2=\langle d\rangle/B=0.072\pm0.013 \tag{4}$$

$$\hat{A}/B^2=\langle E\rangle/B=0.089\pm0.019 \tag{5}$$

$$E_{\max}/B=0.132\pm0.016 \tag{6}$$

In comparison, *Andrews* [1998, p. 175] found $E_{\max}/B\approx0.162$ for his "half-breach" experiments, which used a geometry in which the initial notch was cut at one side of the flume rather than at the center. These dimensionless shape factors presumably reflect details of flow dynamics and fluid/sediment interactions. Analyses such as those of *Parker* [1978], who offered an explanation of the cross-sectional shape of a stable channel, might provide instructive analogies. Here we do not pursue further analysis of the breach shape and simply regard the dimensionless shape factors mentioned above as empirical constraints that influence discharge.

The breach crest is the local high point along streamlines passing through the breach channel and is therefore the location of critical flow. However, the strong curvature of streamlines at the breach crest means that certain relationships valid in steady, uniform, rectilinear flow are no longer correct [*Castro-Orgaz and Chanson*, 2009]: to be specific, the ratio of critical depth d_c to specific energy E no longer equals 2/3, and the critical Froude number is not equal to 1. A regression fit to our experimental data indicates $\langle d\rangle/\langle E\rangle=0.79\pm0.10$, while the Froude number, using the conventional definition [*Chanson*, 2004a, p. 32]

$$\mathrm{Fr}=\frac{Q/A}{\sqrt{gA/B}} \tag{7}$$

has the value 0.74 ± 0.14 (see data tabulated in the SI). Here Q is discharge, calculated using the level-pool approximation

$$Q=Q_{in}-dV_R/dt \tag{8}$$

where Q_{in} is the volumetric recharge rate (usually zero), t is time, and V_R is the reservoir volume as a function of z_{ff}, the water-surface elevation measured far enough from the breach to be outside the region of drawdown. Our calculated values of $\langle d\rangle/\langle E\rangle$ and Fr are consistent with several other studies of strongly convergent flow over weirs: (1) for 2-D flow over cylindrical and round-crested weirs, d_c/E approached 0.8 [*Castro-Orgaz and Chanson*, 2009, p. 776]; (2) Fr = 0.75–0.85 applied for flow over a weir with a trapezoidal profile and narrow crest [*Al-Riffai and Nistor*, 2013; *Al-Riffai*, 2014]; and (3) Fr = 0.78 applied for flow over a

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图8b–8d展示了我们开展的两项实验以及Andrews[1998]开展的一项实验。可见，溃口顶部的形态及水流过水断面形状在时间上几乎保持不变。该过水断面既非梯形，也不类似于工程实践中所采用的任何常规堰型。

h i 括号将用于表示对溃口顶部的平均值，即：

$$\langle F \rangle = (1/B)\int_0^B F\, ds \tag{2}$$

对任意函数 F 成立。溃口顶部水流所占横断面积 A 可表示为 A = hdi B。同样可定义一个假想面积 Â，其代表溃口附近无降落漏斗时的横断面，即

$$\hat{A} = \int_{-\delta s_1}^{(B + ds2} \hat{E}(s)\, ds \tag{3}$$

其中 Ê = zff - zb，ds1、ds2 为微小修正项，代表岸线位置的微小变化。假想水深 Ê 几乎等于 E，即水库相对于溃口顶部的总水头（或比能），其平均值为 ⟨Ê⟩，最大值为 Êmax。Ê 与 E 的微小差异源于溃口顶部流线弯曲这一事实 [Castro-Orgaz 和 Chanson, 2009]。在实际应用中，可忽略 ds1 和 ds2 对 Â 所致的修正，因其远小于由测量误差引起的 Â 不确定度，也远小于 E 与 Ê 的差异；故可简写为 Â ≈ hEi B。根据支撑信息中所列数据，我们得到

$$A/B^2 = hdi/B = 0.072 \pm 0.013$$

$$\hat{A}/B^2 = hEi/B = 0.089 \pm 0.019$$

$$Emax/B = 0.132 \pm 0.016$$

相比之下，Andrews [1998, p. 175] 在其半溃口试验中发现 Emax/B ≈ 0.162；该试验采用的几何构型为初始缺口开在水槽一侧而非中心。这些无量纲形态因子可能反映了流动动力学细节及流体/泥沙相互作用特征。Parker [1978] 等人关于稳定河道横断面形态的分析（曾对此作出解释）或许可提供有益类比。本文未进一步深入分析溃口形态，仅将上述无量纲形态因子视为影响泄流量的经验性约束条件。

溃口顶部是穿过溃口通道流线的局部最高点，因此即为临界流发生位置。然而，溃口顶部流线的强烈曲率意味着稳态、均匀、直线流动下成立的某些关系不再适用 [Castro-Orgaz 和 Chanson, 2009]：具体而言，临界水深 dc 与比能 E 的比值不再等于 2/3，临界弗劳德数亦不等于 1。对本试验数据的回归拟合表明 hdi/hEi = 0.79±0.10；而按常规定义 [Chanson, 2004a, p. 32] 计算的弗劳德数

$$Fr = \frac{Q = A}{\sqrt{(gA/B)}} \tag{7}$$

其值为 0.74±0.14（参见支撑信息中所列数据）。此处 Q 为泄流量，按水平水库近似法计算

$$Q = Qin - dVR/dt \tag{8}$$

其中 Qin 为体积补给率（通常为零），t 为时间，VR 为水库库容，是距溃口足够远处水面高程 zff 的函数（该位置已处于降落漏斗区之外）。我们计算所得的 hdi/hEi 和 Fr 值与其他若干强收敛堰流研究结果一致：(1) 圆柱形及圆顶堰上二维流动中，dc/E 接近 0.8 [Castro-Orgaz 和 Chanson, 2009, p. 776]；(2) 梯形剖面窄顶堰上流动的弗劳德数 Fr = 0.75–0.85 [Al-Riffai 和 Nistor, 2013；Al-Riffai, 2014]；(3) 三角形剖面堰上流动的弗劳德数 Fr = 0.78

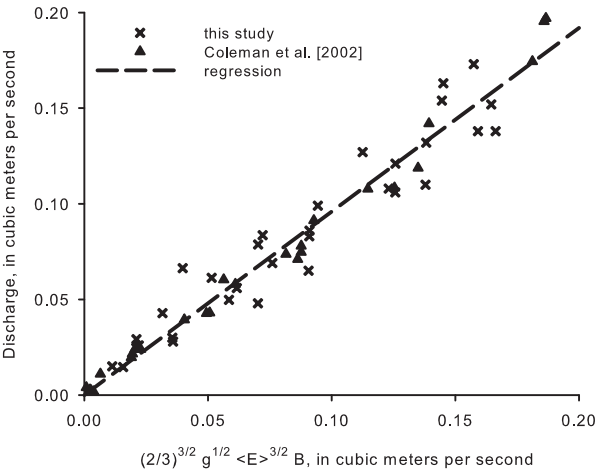


Figure 9. Discharge "rating curve" for the breach crest in both the experiments reported here and those of *Coleman et al.* [2002]. The quantity $(2/3)^{3/2} g^{1/2} \langle E \rangle^{3/2} B$ on the abscissa, where $\langle E \rangle$ is mean specific energy on the breach crest and B is breach crest arc length, makes this analogous to a traditional weir rating curve.

triangular-profile weir, based on data presented by *Kirkgoz et al.* [2008, p. 985] and shown in supporting information Figure S3.

5.4.1. Rating Relationships for Discharge and Possible Dependence on Grain Size

For purposes of computing discharge at the breach crest, the divergence of Froude number and critical depth from their ideal values is represented by incorporating a "discharge coefficient" c_D into a canonical weir equation:

$$Q = c_D \left(\frac{2}{3}\right)^{3/2} g^{1/2} B \langle E \rangle^{3/2} \tag{9}$$

The coefficient $(2/3)^{3/2}$ comes from the analysis of an ideal broad-crested weir [*Chanson*, 2004a, pp. 395–397], which serves as a standard to which other weir shapes are compared. Our data follow the same trend as those of *Coleman et al.* [2002, pp. 835–36], with $c_D = 0.96$ (Figure 9). One may also write "pseudo-weir" relationships that express Q in terms of just $\langle E \rangle$:

$$Q = 5.58\, g^{1/2} \langle E \rangle^{5/2} \tag{10}$$

(see Figure 10) or just B:

$$Q = 0.0126\, g^{1/2} B^{5/2} \tag{11}$$

(see Figure 11). Equations (10) and (11) describe both our data and those of *Coleman et al.* [2002]. Equation (10) would allow introduction of information about the shape of the breach-crest cross section into a 2-D model, while equation (11) raises the intriguing possibility of estimating discharge in real time by measuring B. The close agreement between our results and those of *Coleman et al.* [2002] is striking, especially as the Coleman et al. experiments involved sediments with $D_{50} = 0.5 - 2.4$ mm (considerably coarser than in our experiments) and a constant-head reservoir. To get a better sense of whether the breaching process is independent of grain size, we examined the relationship (Figure 12) between $Q/\sqrt{gD_{50}^5}$ and B/D_{50}, which are dimensionless variables akin to those in *Parker*'s [1979] analysis of hydraulic geometry. The common trend for our data and those of *Coleman et al.* [2002] is

$$\log\left(\frac{Q}{g^{1/2} D_{50}^{5/2}}\right) = -1.34 + 2.36 \log\left(\frac{B}{D_{50}}\right) \tag{12}$$

which is equivalent to

$$Q = 0.046\, g^{1/2} B^{5/2} \left(\frac{D_{50}}{B}\right)^{0.14} \tag{13}$$

This slight increase in Q with increasing D_{50} contrasts with results of *Hakimzadeh et al.* [2013], who found a slight but consistent decrease in peak discharge as D_{50} was increased from 0.25 to 2 mm in dam-breaching experiments that differed only in the choice of grain size. An inverse

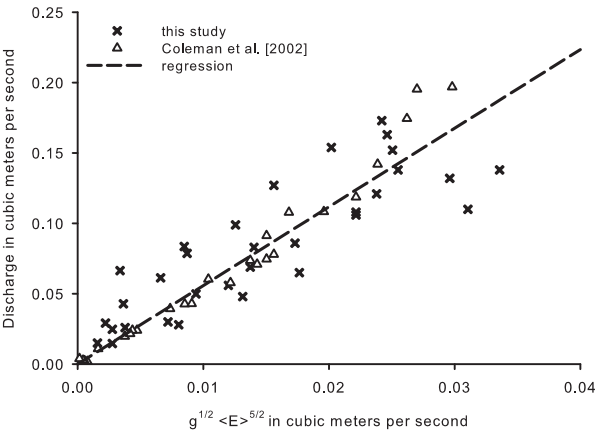


Figure 10. A pseudo-rating curve showing discharge as a function of the mean specific energy on the breach crest, $\langle E \rangle$, for experiments reported here as well as those of *Coleman et al.* [2002]. The relationship between Q and $\langle E \rangle$ could be incorporated as a constraint in a parametric model of breaching.

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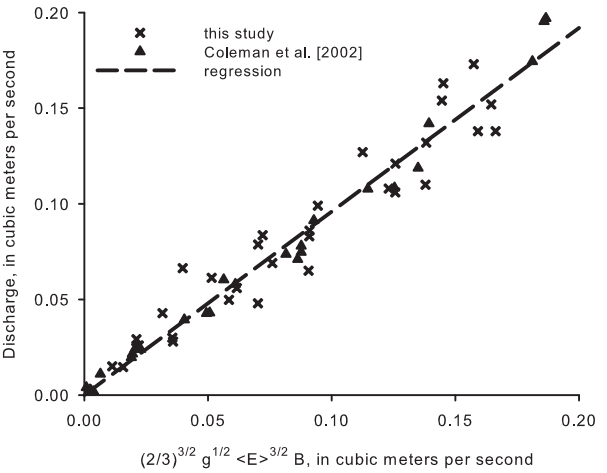


图9. 本研究及 Coleman 等人 [2002] 所报道试验中溃口顶部的泄流关系曲线。横坐标为 $(2/3)^{3/2}\ g^{1/2}\ hEi^{3/2}\ B$，其中 hEi 为溃口顶部平均比能，B 为溃口顶部弧长；由此使其类同于传统堰流关系曲线。

三角形剖面堰，依据 Kirkgoz 等人 [2008, p. 985] 提供的数据绘制，详见支撑信息图 S3。

5.4.1. 泄流关系式及其对颗粒粒径的可能依赖性

为计算溃口顶部泄流量，弗劳德数与临界水深偏离其理想值的现象，可通过在标准堰流方程中引入泄流系数 cD 加以表征：

$$Q = cD\ \frac{(2/3)^{3/2}\ g^{1/2}\ B\ hEi^{3/2}}{2} \tag{9}$$

系数 $(2/3)^{3/2}$ 来源于对理想宽顶堰的分析 [Chanson, 2004a, pp. 395–397]，该分析为其他各类堰型提供了标准参照。本研究数据与 Coleman 等人 [2002, pp. 835–36]，其中 cD = 0.96（图9）。也可写出仅以 hEi 表达 Q 的伪堰流关系式：

$$Q = 5.58\ g^{1/2}\ hEi^{5/2} \tag{10}$$

（见图10）或仅与 B 相关：

$$Q = 0.0126\ g^{1/2}\ B^{5/2} \tag{11}$$

（见图11）。公式(10)和(11)均能同时描述本研究数据与 Coleman 等人 [2002] 的数据。公式(10)可将溃口顶部横断面形状信息引入二维模型；而公式(11)则提出了通过实测 B 值实时估算泄流量的有趣可能性。本研究结果与 Coleman 等人 [2002] 结果之间高度一致，尤为显著的是：Coleman 等人的试验所用沉积物 D50 = 0.5–2.4 mm（明显粗于本研究），且采用恒定水头水库。为更深入理解溃决过程是否独立于颗粒粒径，我们考察了 Q =

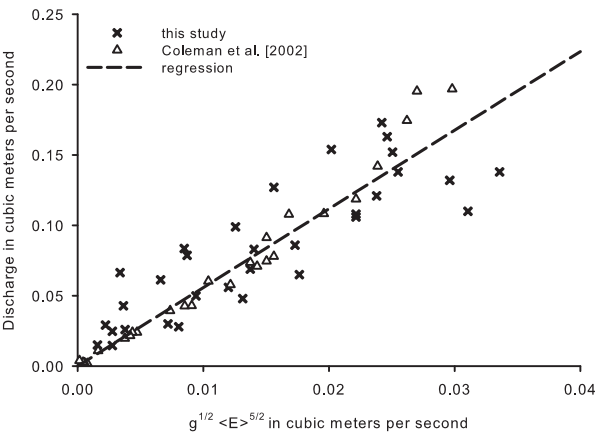


图10. 一条伪泄流关系曲线，显示本研究及 Coleman 等人 [2002] 所报道试验中溃口顶部平均比能 hEi 与泄流量 Q 之间的关系。Q 与 hEi 之间的关系可作为约束条件纳入溃口发展的参数化模型中。

$\sqrt{g\ D^5}$ 以及 B/D50，二者均为无量纲变量，类似于 Parker [1979] 对水力几何形态所作分析中的变量。本研究数据与 Coleman 等人 [2002] 数据共同呈现的趋势是

$$\log\left(\frac{Q}{g^{1/2}\ D^{5/2}}\right) = -1.34 + 2.36\log\left(\frac{B}{D_{50}}\right) \tag{12}$$

该式等价于 $Q = 0.046\ g^{1/2}$

$B^{5/2}\ D50$ $\left(\frac{}{B}\right)^{0.14}$ (13)

Q 随 D50 增大而出现的这一微小增加，与 Hakimzadeh 等人 [2013] 的结果形成对比；后者在仅改变颗粒粒径的坝体溃决试验中发现，当 D50 从 0.25 mm 增至 2 mm 时，峰值流量出现微小但一致的下降。一种反向关系

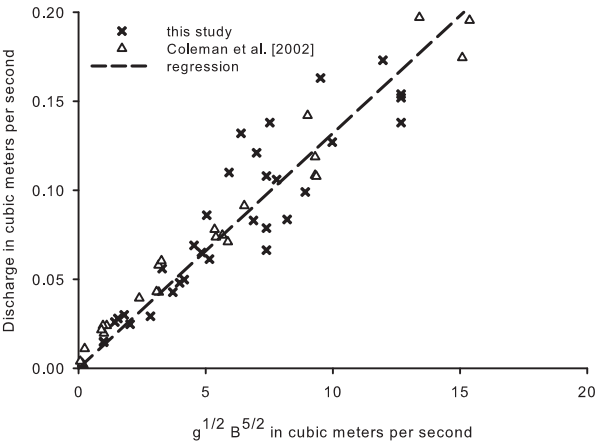


Figure 11. A pseudo-rating curve showing discharge as a function of the length of the breach crest, *B*, for the experiments reported here as well as those of *Coleman et al.* [2002].

relationship between Q_{peak} and D_{50} was also reported by *Schmocker and Hager* [2012] for 2-D experiments—that is, with the dam overtopped from one flume wall to the other—with constant inflow to the reservoir. Whether discharge depends on grain size thus remains ambiguous, although it appears that if such a dependence does exist, it is a weak one.

We also explored the possibility of using $\tilde{B}$, the straight-line distance between the ends of the breach crest (Figure 2), as a proxy for *B*. $\tilde{B}$ is relatively easy to determine by simply making measurements on photos taken from overhead, as long as one accounts for optical distortion as one moves away from $y=0$. $\tilde{B}$ may be thought of as a measure of breach-channel width, but unlike width measured at the dam crest, which is shaped primarily by episodic bank failures [cf. *Hunt et al.*, 2005], $\tilde{B}$ reflects primarily tractional erosion with rare perturbations by slope failures. (This is readily confirmed by carefully watching the Movies.) $\tilde{B}$ and *Q* are well correlated, but with a complication—hysteresis (Figure 13)—that is unsurprising: the breach crest is, after all, not an equilibrium morphological feature: it continues to migrate upstream even after discharge begins to decrease. *Al-Riffai* [2014, chapter 6] also measured $\tilde{B}$ during his breaching experiments and found that $B/\tilde{B}$ ranged from about 1.05 to 1.45. We found practically the same range of values (see the supporting information).

5.5. Hydrographs, Reproducibility, and the Role of Groundwater

Al-Riffai [2014] has recently shown that using very careful construction methods—controlling water content, using thin sediment layers, and applying a uniform compaction effort to the surface of each layer—it is possible to do replicate breaching experiments with sand dams that yield hydrographs that are nearly identical in timing as well as shape. We also did near-replicate experiments (Table 1), albeit at a much larger scale than Al-Riffai, without controlling water content and with a simple method of compaction. We first consider our six experiments with $Z_0 = 0.98$ or 0.99 m (Figure 14a). While the hydrographs for these experiments are clearly shaped similarly, they do not overlap. The same six hydrographs are shown in Figure 14b using a different time datum: instead of taking $t=0$ as the time when the notch in the dam was cut, $t=0$ was chosen as the time when a retreating step, or headcut, intersected the breach crest—that is, the start of stage III (Table 3). The six hydrographs then come into much closer alignment. The same sort of hydrograph comparison is shown in Figures 14c–14e for pairs of replicate experiments with, respectively, $Z_0 \approx 0.87$ m, $Z_0 \approx 0.76$ m, and $Z_0 \approx 0.55$ m; again one sees that measuring time relative to the start of stage III brings the hydrographs into alignment. Clearly, then, the variation between "unshifted" hydrographs (i.e., with $t=0$ at the time the notch is cut) is almost entirely a reflection of experiment-to-experiment differences in breach evolution during stages I and II, when the stream is incising unsaturated sand near the downstream dam face until it finally reaches the water table within the

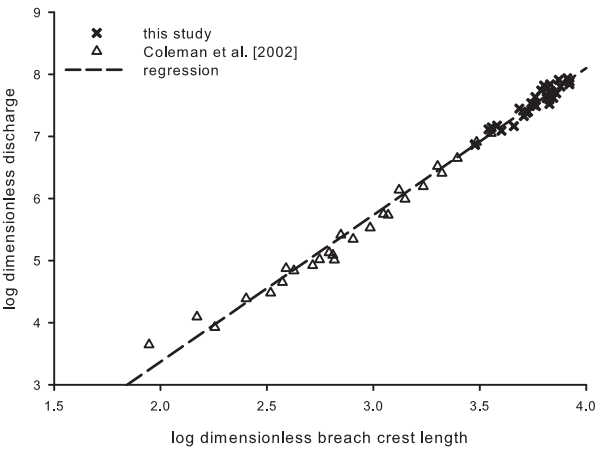


Figure 12. A log-log plot of nondimensional discharge ($Q/\sqrt{g D_{50}^5}$) as a function of nondimensional breach crest length (B/D_{50}), for the experiments reported here as well as those of *Coleman et al.* [2002] The nondimensional variables were chosen following *Parker* [1979].

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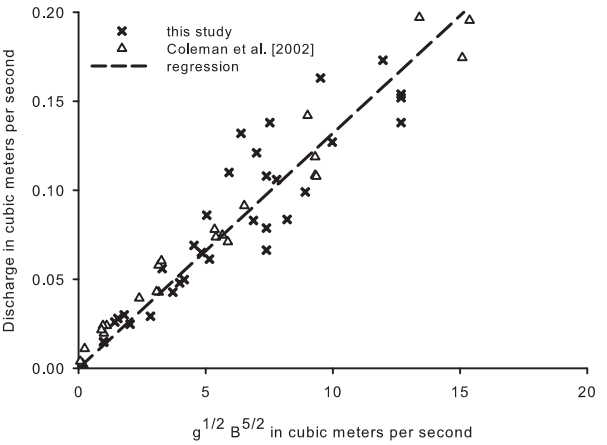


图11. 假定率定曲线，显示本文实验及Coleman等[2002]实验中泄流量Q随溃口顶部长度B的变化关系。

Schmocker与Hager [2012]亦曾针对二维实验（即坝体自一个水槽壁溢流至另一侧）报告了Q_{ea}与D的关系，其实验中水库进水流量恒定。因此，泄流量是否依赖于颗粒尺寸仍存疑问，尽管若有此依赖性，其影响亦属微弱。

我们还探讨了采用~B（即溃口顶部两端点间的直线距离，见图2）作为B的替代指标的可能性。~B相对易于确定，只需对俯拍照片进行测量，并在远离y=0处校正光学畸变即可。~B可视为溃口河道宽度的一种度量，但不同于宽度

在坝顶测得，主要受间歇性岸坡失稳控制 [参见Hunt等, 2005]，~B则主要反映牵引侵蚀，偶有坡面失稳扰动。（通过仔细观看视频可轻易证实。）~B与Q相关性良好，但存在滞后现象（见图13），这并不意外：溃口顶部毕竟并非均衡地貌特征，即使泄流量开始减小后，其仍持续向上游迁移。Al-Riffai [2014, 第6章] 在其溃坝实验中亦测定了~B，发现B/~B比值介于约1.05至1.45之间。我们得到的实际比值范围与此基本一致（见支撑信息）。

5.5 水文过程线、可重复性及地下水作用

Al-Riffai [2014]最近表明，采用极为精细的筑坝工艺——精确控制含水量、铺设薄层沉积物、并对每层表面施加均匀压实功——可实现砂质坝体溃决实验的重复性，所得水文过程线在时序与形态上均高度一致。我们亦开展了近似重复实验（见表1），但规模远大于Al-Riffai，且未控制含水量，仅采用简易压实方法。我们首先考察Z=0.98或0.99 m的六次实验（见图14a）。这些实验的水文过程线形态明显相似，但并未重合。图14b以不同时间基准展示相同六条水文过程线：此处t并非取为坝体开槽时刻，而是取为

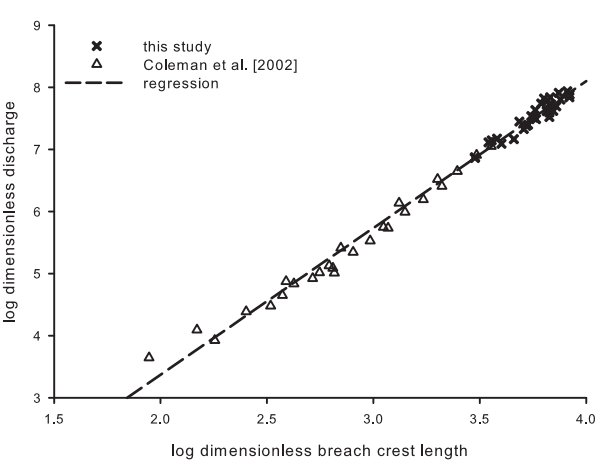


图12. 本文实验及Coleman等[2002]实验的无量纲泄流量（$Q=p\sqrt{g}$作为无量纲溃口顶部长度（B/D）的函数，涵盖本文实验及Coleman等[2002]的实验。无量纲变量依据Parker [1979]选定。

即第三阶段起始时刻（见表3）。此后，六条水文过程线高度趋同。图14c–14e同样展示了溃口初始高度Z≈0.87 m、Z≈0.76 m及Z≈0.55 m的重复实验对的水文过程线比较；同样可见，以第三阶段起始时刻为时间基准，水文过程线趋于一致。显然，未经时间校正的水文过程线（即以开槽时刻为t）之间的差异，几乎完全反映了各次实验在第一、二阶段溃口演化过程的差异——此阶段河道在下游坝面附近未饱和砂层中下切，直至最终抵达坝体内潜水面。

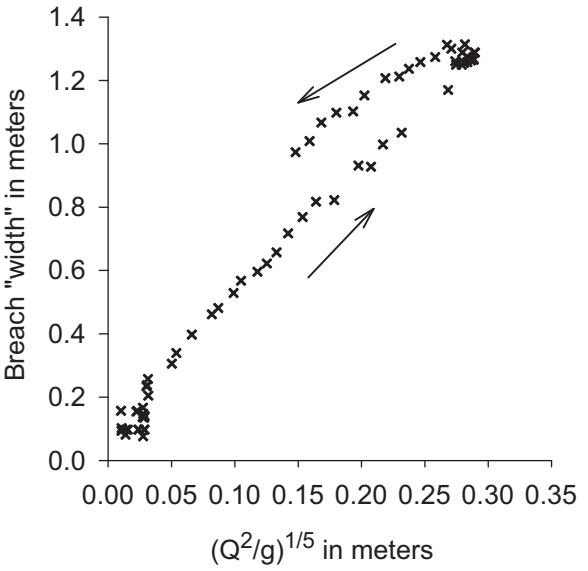


Figure 13. Hysteresis in the measured relationship between discharge and breach "width" $\tilde{B}$ in Experiment 3. Arrows indicate the sense of the change in discharge with time.

dam. Thus we consider the groundwater hydrology and its relationship to stream incision.

While a complete exploration of the groundwater hydrology is beyond the scope of this paper, we can gain valuable insight by modeling and comparing the groundwater system for two experiments that involved different degrees of infiltration from the reservoir: namely, Experiments 4 and 5. Experiment 4 had the largest discharge of any of the six experiments represented in Figure 14a. The computational method (described in supporting information Text S2) uses the equations of variably saturated flow to solve for the evolving groundwater state of the dams as water infiltrates from the reservoir. Figure 15 shows modeled distributions (constrained by measurements) of pressure head ψ within the dams at the time the notch was cut in the dam crest. These distributions are quite close to steady state and may therefore be regarded as applying throughout stages I and II. For both dams, pressure head ranges from the initial value of -0.4 m (in the driest regions near the downstream dam face) to about 0.9 m beneath the reservoir. The position of the water table (the $\psi=0$ contour) is controlled by the position of the drain and the reservoir level.

Now envisage an incipient channel just barely incised into the downstream dam face. Along this entire reach, water will tend to infiltrate from the channel into the partially saturated sand beneath. While one could calculate the infiltration rate [cf. *Al-Riffai*, 2014], it is adequate for present purposes to note that the maximum rate of advance of the wetting front is limited by the saturated hydraulic conductivity K_{sat}, measured at close to 10^{-4} m/s (Table 2). (We implicitly use the Green-Ampt approximation for infiltration—see *Smith* [2002, pp. 34–36].) If the erosion rate exceeds K_{sat}, then infiltration is negligible and the channel incises sediment whose hydrologic state is as shown in Figure 15. Although we do not have direct measurements of erosion rate along the downstream dam surface, it is possible to make rough estimates by viewing videos and estimating the rate at which the surface of the stream is dropping relative to the original dam surface. We find that during stage I and the early part of stage II, erosion rate along the lower half of the downstream dam face was about 5 to 10 mm/s, or two orders of magnitude greater than the maximum infiltration rate. Thus infiltration into the sand beneath the stream is in fact negligible and we can ignore the effect of seepage on erosion [cf., e.g., *Liu and Chiew*, 2012].

Comparing Figures 15a and 15b, we see that the extent of the partially saturated zone is significant greater in Experiment 5 than in Experiment 4. We would therefore expect that the time required to erode down to the water table should be greater in Experiment 5 than in Experiment 4. Once the water table is reached, however, the sand becomes easily eroded regardless of the state of the groundwater system at the beginning of an experiment. Thus we expect the onset of rapid erosion, the formation of a stepped stream profile, and the retreat of a headcut to the breach crest—which marks the beginning of stage III—to occur sooner in Experiment 4 than in Experiment 5. The sharp increase in Q should likewise occur sooner in Experiment 4 than in Experiment 5. These expectations are borne out by the measured hydrographs.

There is another consequence of choosing $t=0$ as the start of stage III: namely, the time to peak discharge (t_p) turns out to vary little from experiment to experiment (Table 3 and Figure 16). This result differs from that of *Al-Riffai* [2014, pp. 242–252], who found that t_p varied in accord with so-called Froude scaling [*Hughes*, 1993, pp. 69–70], such that if all length scales of a physical model are multiplied by a factor of λ relative to a prototype, t_p scales with $\sqrt{\lambda}$. The reason for the difference between our data and Al-Riffai's is as yet unclear.

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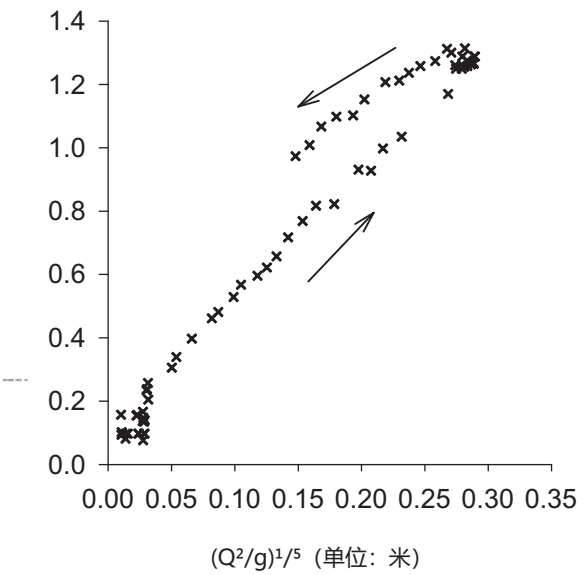


图13. 实验3中实测泄流量Q与溃口宽度~B关系的滞后现象。箭头表示随时间推移泄流量变化的方向。

坝体。因此，我们需考察地下水水文学及其与河道下切的关系。

尽管全面探讨地下水水文学已超出本文范畴，但我们可通过建模并比较两次入渗程度不同的实验（即实验4与实验5）的地下水系统，获得宝贵见解。实验4是图14a所示六次实验中泄流量最大的一次。所用计算方法（见支撑信息文本S2）基于非饱和—饱和水流方程，求解水库水入渗过程中坝体地下水状态的演化过程。图15展示了在坝顶开槽时刻，经实测约束的坝体内压力水头w的模拟分布。这些分布已非常接近稳态，因而

可视为在整个第一、二阶段均适用。对于两座坝体，压力水头范围从下游坝面附近最干燥区域的初始值20.4 m变化至水库下方约0.9 m。潜水面位置（即w=0.5等值线）由排水口位置及水库水位控制。

现设想一条初生河道，仅轻微切入下游坝坡表面。沿该河段全长，水流将倾向于从河道渗入其下方的部分饱和砂层。尽管可计算入渗速率 [参见Al-Riffai, 2014]，但就当前目的而言，只需指出湿润锋最大推进速率受饱和水力传导度Ksat限制，其测量值约为10^4 m/s（见表2）。（我们隐式采用Green-Ampt入渗近似法——参见Smith [2002, 第34–36页]。）若侵蚀速率超过Ksat，则入渗可忽略不计，河道下切的沉积物水文状态如图15所示。虽然我们未直接测定下游坝面的侵蚀速率，但可通过观看实验视频并估算河道水面相对于原始坝面的下降速率，进行粗略估计。我们发现，在第一阶段及第二阶段早期，下游坝面下半部分的侵蚀速率约为5至10 mm/s，比最大入渗速率高两个数量级。因此，河道下方砂层的入渗实际可忽略不计，我们可忽略渗流对侵蚀的影响 [参见，例如Liu和Chiew, 2012]。

对比图15a与图15b可见，实验5中部分饱和带范围显著大于实验4。因此我们预计，实验5中下切至潜水面所需时间应长于实验4。然而，一旦触及潜水面，无论实验初始时地下水系统处于何种状态，砂土均易于侵蚀。故而我们预期，快速侵蚀的起始、阶梯状河道纵剖面的形成以及溯源侵蚀前锋退至溃口顶部（即第三阶段起始标志）在实验4中应早于实验5发生。Q值的急剧上升同样应在实验4中早于实验5出现。实测水文过程线验证了上述预期。以t50作为第三阶段起始点还有另一后果：即峰值流量出现时间（tp）在各次实验间变化甚微（见表3与图16）。该结果与Al-Riffai [2014, 第242–252页] 的发现不同，后者发现tp遵循所谓弗劳德相似律 [Hughes, 1993, 第69–70页]，即若物理模型所有长度尺度相对于原型按比例因子k缩放，则tp按

$\sqrt{\lambda}$。我们的数据与Al-Riffais结果差异的原因在于尚不明确。

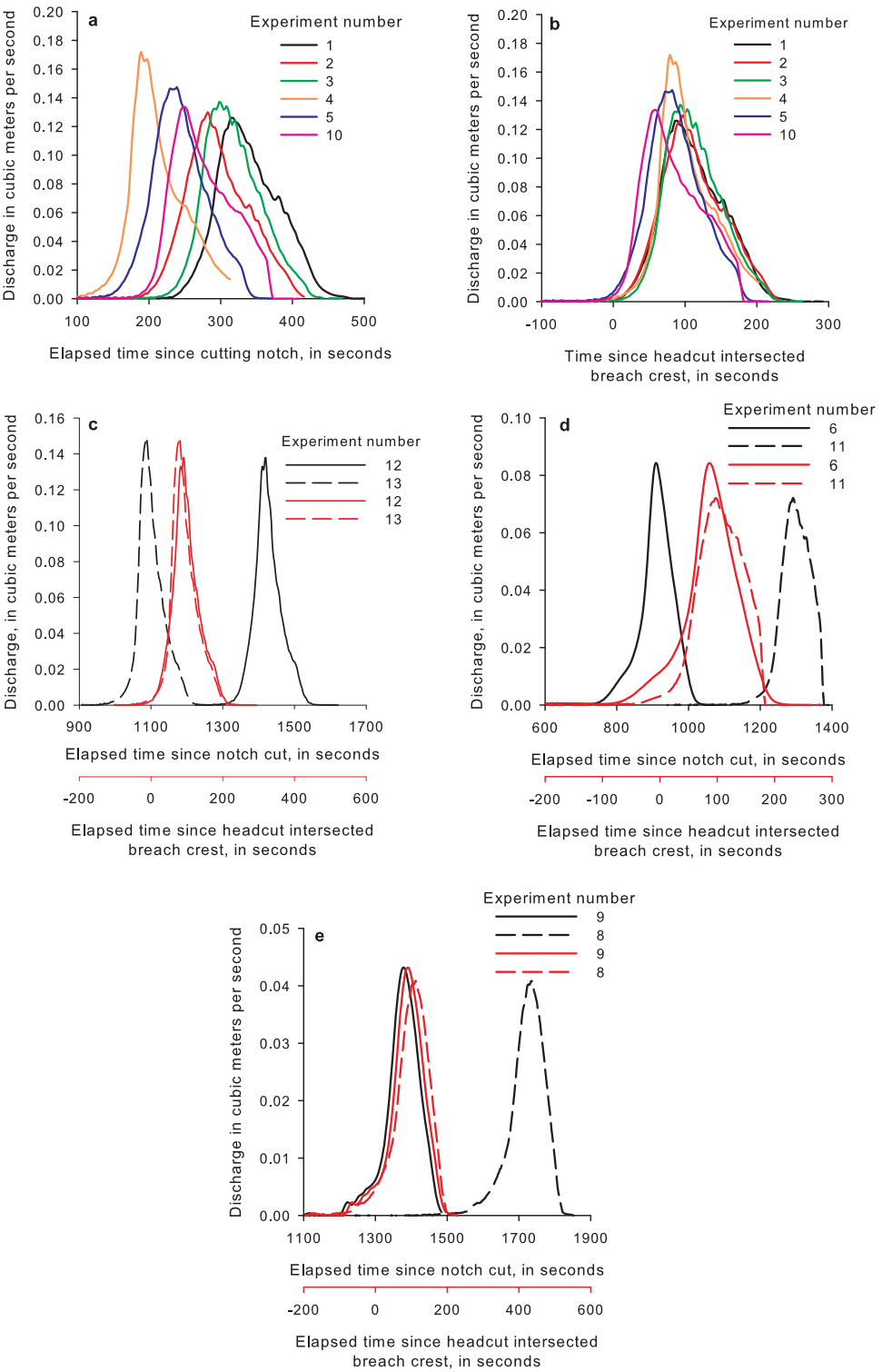


Figure 14. Hydrographs for replicate dam-breach experiments reported in this paper, with alternate choices of the time datum. (a) Experiments with initial dam heights close to 1 m. Times are measured relative to the moment when the notch was cut across the dam crest. (b) Experiments with initial dam heights close to 1 m, but with time measured relative to the beginning of morphological stage III. (c) Experiments with initial dam heights close to 0.87 m. Black curves (and time axis) apply when time is measured relative to the moment that the notch was cut across the dam crest. Red curves (and offset time axis) apply when time is measured relative to the beginning of morphological stage III. (d) Experiments with initial dam heights close to 0.76 m. Black curves (and time axis) apply when time is measured relative to the moment that the notch was cut across the dam crest. Red curves (and offset time axis) apply when time is measured relative to the beginning of morphological stage III. (e) Experiments with initial dam heights close to 0.55 m. Black curves (and time axis) apply when time is measured relative to the moment that the notch was cut across the dam crest. Red curves (and offset time axis) apply when time is measured relative to the beginning of morphological stage III.

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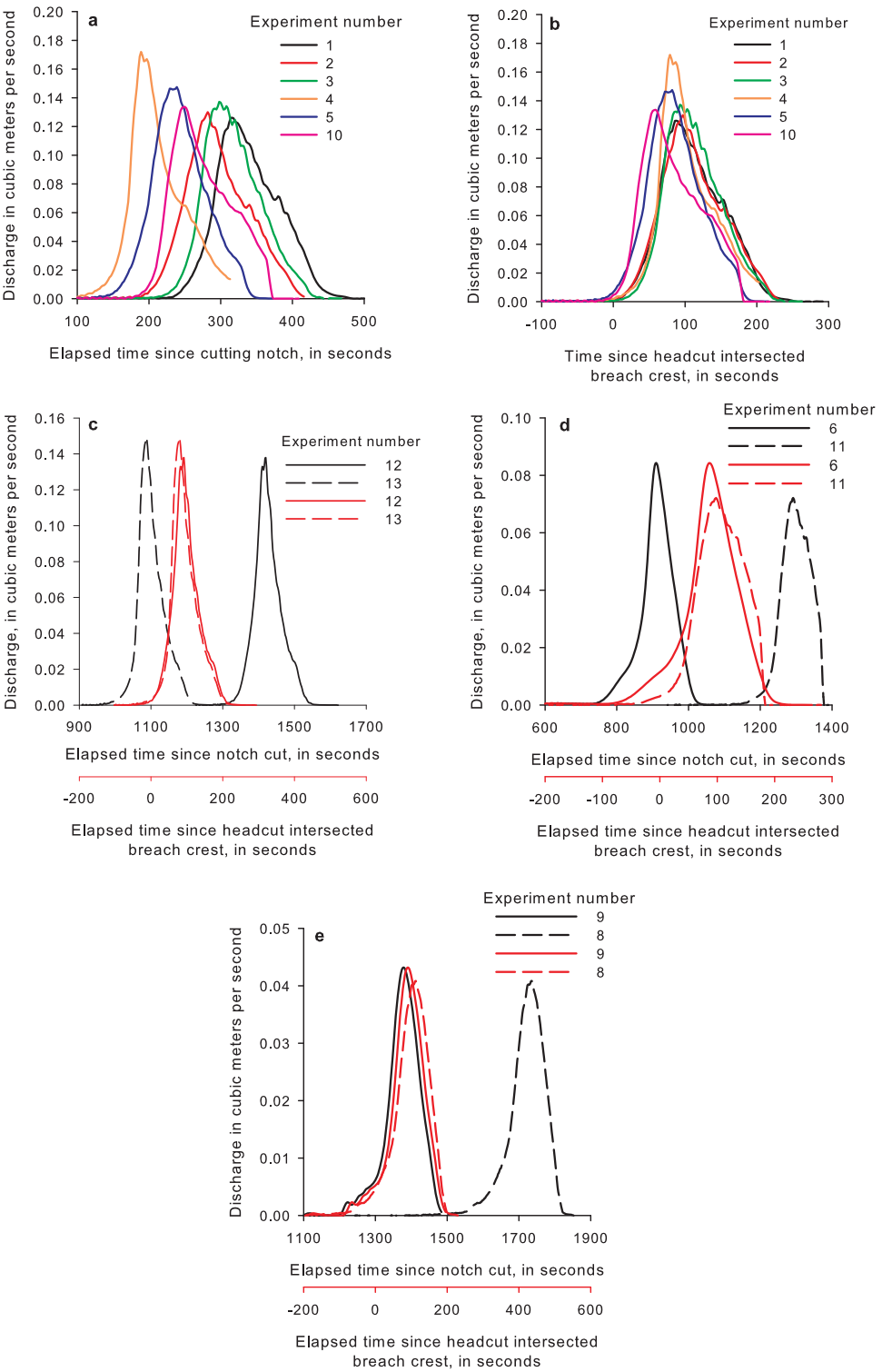


图14. 本文报道的重复溃坝实验水文过程线，采用不同时间基准。（a）初始坝高接近1米的实验。时间以在坝顶开槽时刻为起点。（b）初始坝高接近1米的实验，但时间以地貌第三阶段起始时刻为起点。（c）初始坝高接近0.87米的实验。黑色曲线（及时间轴）对应以在坝顶开槽时刻为起点的时间；红色曲线（及偏移时间轴）对应以地貌第三阶段起始时刻为起点的时间。（d）初始坝高接近0.76米的实验。黑色曲线（及时间轴）对应以在坝顶开槽时刻为起点的时间；红色曲线（及偏移时间轴）对应以地貌第三阶段起始时刻为起点的时间。（e）初始坝高接近0.55米的实验。黑色曲线（及时间轴）对应以在坝顶开槽时刻为起点的时间；红色曲线（及偏移时间轴）对应以地貌第三阶段起始时刻为起点的时间。

Table 3. Timing of Key Events During Experiments

Experiment[a]	Δt_{res}[b]	t_n[c]	t_t[d]	t_{hc}[e]	t_p[f]	Δt_{tn}[g]	Δt_{hcn}[h]	Δt_{pn}[i]	Δt_{pt}[j]	Δt_{phc}[k]
1	3235	3452	3633	3680	3768	181	228	316	135	88
2	3146	3348	3509	3533	3630	161	185	282	121	97
3	2585	2810	3003	3015	3108	193	205	298	105	93
4	2695	4384	4465	4494	4573	81	110	189	108	79
5	1100	2300	2414	2457	2539	114	157	239	125	82
6	1190	3825	4409	4648	4750	584	823	925	341	102
8	420	6454	7345	8077	8189	891	1623	1735	844	112
9	350	5732	6468	7020	7110	736	1288	1378	642	90
10	1030	3074	3211	3265	3324	137	191	260	113	59
11	597	4821	5745	6015	6112	924	1194	1291	367	97
12	773	3160	4380	4488	4578	1220	1328	1418	198	90
13	723	4618	5522	5625	5706	904	1007	1088	184	81
14	622	5361		8415	8542		3054	3181		127

[a]Experiment 7 was run with a layer of gravel and cobbles within the dam and is not discussed in this paper.
[b]Δt_{res} is period of time from start of data collection until the reservoir is filled to the desired level, in seconds.
[c]t_n is time when final adjustment is made to notch, in seconds relative to start of data logging.
[d]t_t is time when incipient channel reaches toe of dam, in seconds relative to start of data logging.
[e]t_{hc} is time when headcut intersects breach crest, in seconds relative to start of data logging. This is the beginning of morphological stage III.
[f]t_p is time of peak discharge, in seconds relative to start of data logging.
[g]Δt_{tn} is interval of time between final notch adjustment and incipient channel reaching toe of dam, in seconds.
[h]Δt_{hcn} is interval of time between final notch adjustment and headcut intersecting breach crest, in seconds.
[i]Δt_{pn} is interval of time between final notch adjustment and peak discharge, in seconds.
[j]Δt_{pt} is interval of time between incipient channel reaching toe of dam and peak discharge, in seconds.
[k]Δt_{phc} is interval, of time between headcut intersecting breach crest and peak discharge, in seconds.

5.6. Dependence of Peak Discharge on Initial Water Level

In addition to our experiments reported here, two other studies have included systematic variation of dam height, and thus of initial water level: *Al-Riffai* [2014, chapter 6] and *Hakimzadeh et al*. [2013]. Al-Riffai reported results of so-called scale-series breaching experiments in which dam geometry was varied systematically: In some experiments, both the length and height of the dam were scaled by a factor λ (equal

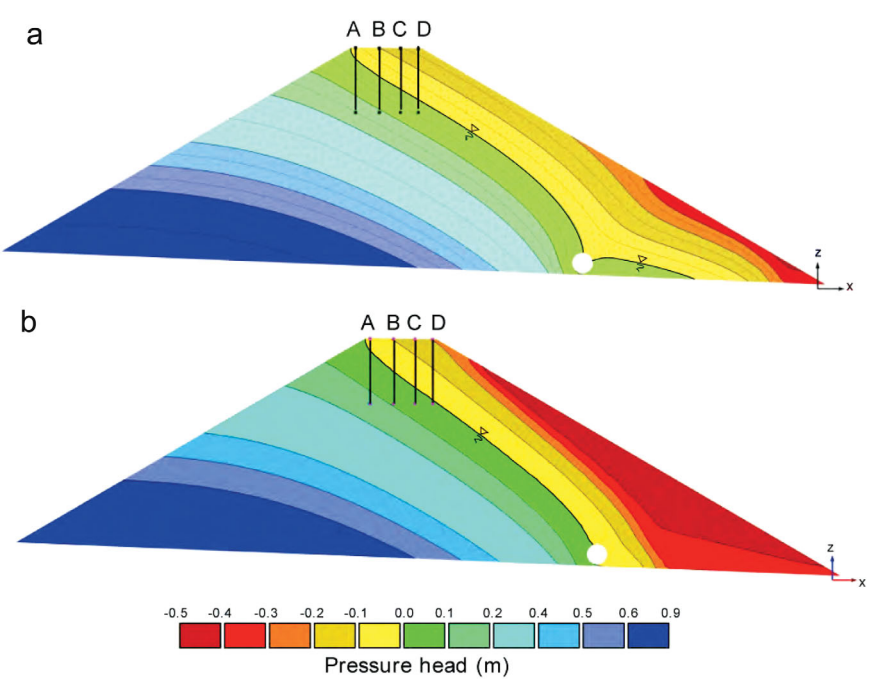


Figure 15. Modeled distributions of pressure head within dams just before the start of breaching, as constrained by piezometer and tensiometer measurements. These are very close to steady state head distributions. (a) Experiment 4 (b) Experiment 5. A more detailed discussion of these model results is found in the supporting information.

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表3. 实验期间关键事件的时间安排

	Δt_{res} [b]	t_n [c]	t_t [d]	t_{hc} [e]	t_p [f]	Δt_{tn} [g]	Δt_{hcn} [h]			
1	3235	3452	3633	3680	3768	181	228	316	135	88
2	3146	3348	3509	3533	3630	161	185	282	121	97
3	2585	2810	3003	3015	3108	193	205	298	105	93
4	2695	4384	4465	4494	4573	81	110	189	108	79
5	1100	2300	2414	2457	2539	114	157	239	125	82
6	1190	3825	4409	4648	4750	584	823	925	341	102
8	420	6454	7345	8077	8189	891	1623	1735	844	112
9	350	5732	6468	7020	7110	736	1288	1378	642	90
10	1030	3074	3211	3265	3324	137	191	260	113	59
11	597	4821	5745	6015	6112	924	1194	1291	367	97
12	773	3160	4380	4488	4578	1220	1328	1418	198	90
13	723	4618	5522	5625	5706	904	1007	1088	184	81
14	622	5361		8415	8542		3054	3181		127

a 实验7在坝体内铺设了一层砾石与卵石，本文未予讨论。
b Δt_{res}为数据采集起始至水库充水至预定水位所需时间（秒）。
c t_n为最终缺口调整完成时刻（以数据记录起始时刻为零点，单位：秒）。
d t_t为初始河道抵达坝踵时刻（以数据记录起始时刻为零点，单位：秒）。
e t_{hc}为头坎与溃口顶部相交时刻（以数据记录起始时刻为零点，单位：秒）。此即地貌第三阶段的起始时刻。
f t_p为洪峰流量出现时刻（以数据记录起始时刻为零点，单位：秒）。
g Δt_{tn}为最终缺口调整完成至初始河道抵达坝踵的时间间隔（秒）。
h Δt_{hcn}为最终缺口调整完成至头坎与溃口顶部相交的时间间隔（秒）。
i Δt_{p n}为最终缺口调整完成至洪峰流量出现的时间间隔（秒）。
j Δt_{pt}为初始河道抵达坝踵至洪峰流量出现的时间间隔（秒）。
k Δt_{phc}为头坎与溃口顶部相交至洪峰流量出现的时间间隔（秒）。

5.6. 洪峰流量对初始水位的依赖性

除本研究外，另有两项研究系统性地改变了坝高（即初始水位）：Al-Riffai[2014，第6章]及Hakimzadeh等[2013]。Al-Riffai报告了所谓“比例系列溃口实验”结果，其中坝体几何尺寸被系统性调整：在部分实验中，坝长与坝高均按比例系数k缩放（k等于……

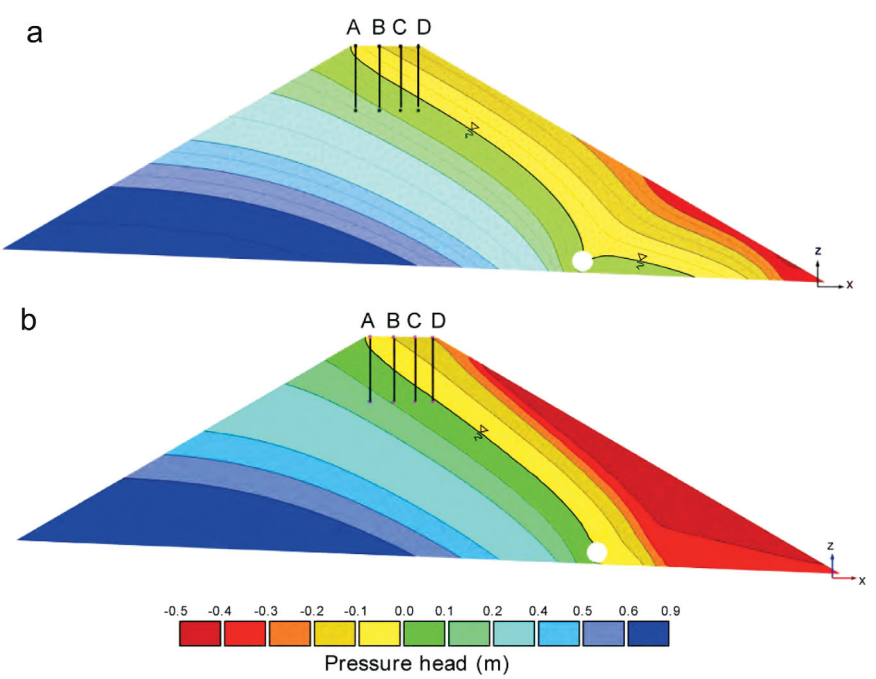


图15. 溃口发生前坝体内压力水头的模拟分布图，该模拟结果受测压计与张力计实测数据约束。这些分布非常接近稳态水头分布。（a）实验4；（b）实验5。有关这些模拟结果的更详细讨论见支撑信息。

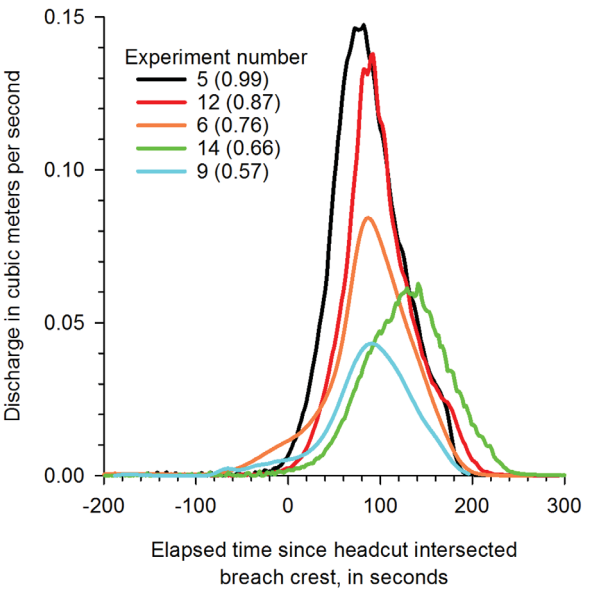


Figure 16. Comparison of hydrographs for five of our experiments, each one conducted with a different initial water level (in meters, given in parentheses). Time datum is the beginning of morphological stage III. Note that while there is systematic variation in peak discharge, there is little variability in the *time* to peak discharge.

to either 1/2 or 1/3) relative to a benchmark experiment with a 0.3 m high dam, while in other experiments, only the length was scaled by λ. Hakimzadeh et al. worked with two different cohesionless sediments (D_{50} = 0.25 or 2.0 mm) and simply used an initial water level of either 0.3 or 0.4 m. Peak discharge for all of these experiments, as a function of initial water level, is shown in Figure 17. The data have been normalized so that all results can be conveniently shown on a single graph. Although the data are limited, results from all three sets of experiments are broadly consistent: a general trend of Q_p decreasing with initial water level. (Note that in contrast to Al-Riffai's experiments, neither ours nor those of Hakimzadeh et al. can be regarded as constituting scale series, as both dam height and dam shape were varied.) The fact that such a trend exists is not a trivial observation: after all, the eroding breach crest does not "know" the initial water depth. Put more formally, one could plausibly argue that as long as flow depth at the breach crest is small compared to Z_0, the hydraulics at the breach crest, and thus the local shear stress, should be independent of Z_0. If the rate of breach crest erosion depends upon local shear stress, then the breach hydrograph should also be independent of Z_0. As this is demonstrably incorrect, at least one of the assumptions made must be incorrect.

One possible resolution of the conundrum is that erosion rate at the breach crest is not a function of local shear stress, but rather is constrained by the fact that the headcut just downstream of the breach crest is maintained at the angle of static friction, ϕ. Consider Figure 18, which shows a simplified 2-D picture of flow through the breach channel. Water pouring over the breach crest (average elevation z_c) will erode the

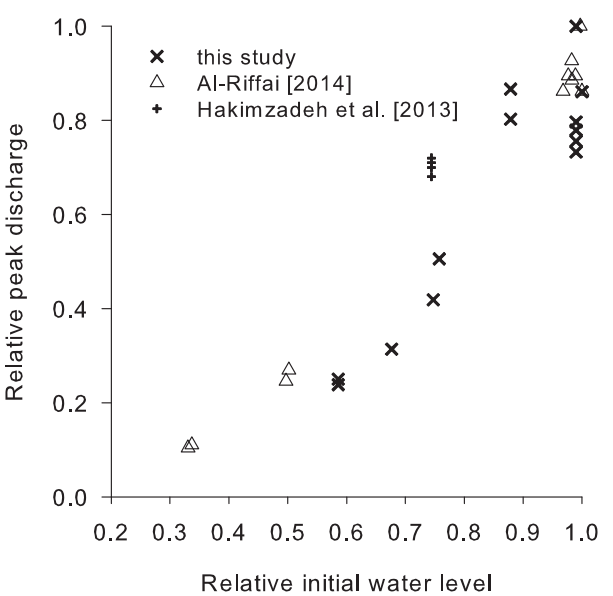


Figure 17. Graph of nondimensional peak discharge as a function of nondimensional initial water level, for our data and data of *Al-Riffai* [2014] and *Hakimzadeh et al.* [2013]. For each data set, the scale for water level is the maximum used; the scale for peak discharge is the maximum found for the tallest dam in the data set. All data sets show a decrease in peak discharge as initial water level decreases.

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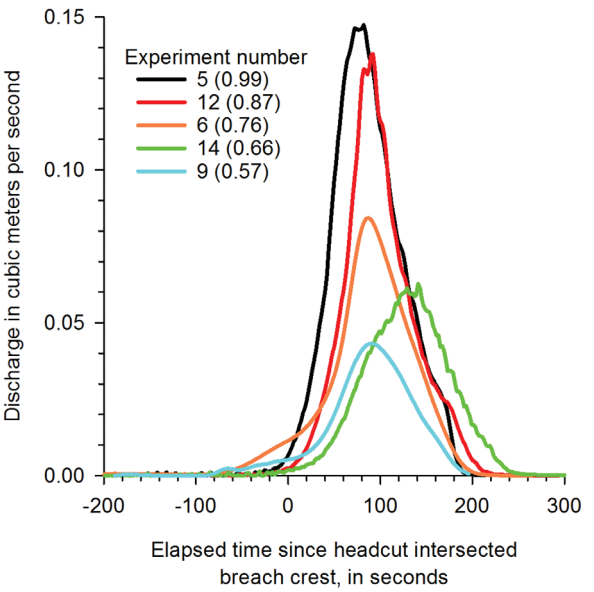


图16. 本研究五组实验的洪水过程线对比图，各实验初始水位（单位：米）不同（括号内数值）。时间零点为地貌第三阶段起始时刻。可见洪峰流量存在系统性变化，但洪峰到达时间变异性甚小。

.....相对于一座0.3米高坝的基准实验，高度与长度均按比例系数k缩放（k取1/2或1/3）；而在其他实验中，仅长度按k缩放。Hakimzadeh等采用两种不同无黏性沉积物（D₅₀=0.25毫米或2.0毫米），初始水位则简单设定为0.3米或0.4米。所有这些实验的洪峰流量随初始水位变化关系如图17所示。数据经归一化处理，以便于在同一张图中直观展示全部结果。尽管数据有限，但三组实验结果总体一致：Q_p随初始水位升高而增大。（需注意：与Al-Riffai的实验不同，我们的实验及Hakimzadeh等的实验均不能视为严格的比例系列实验，因其坝高与坝形均有所变化。）此类趋势的存在绝非平凡观察：毕竟，正在侵蚀的溃口顶部本身并不“知晓”初始水深。更严谨地说，人们可合理推断：只要.....

只要溃口顶部水深远小于Z₀，则溃口顶部的水力特性（从而局部剪应力）应与Z₀无关。若溃口顶部侵蚀速率取决于局部剪应力，则溃口洪水过程线亦应与Z₀无关。但事实显然并非如此，故前述假设中至少有一项不成立。

该难题的一种可能解释是：溃口顶部的侵蚀速率并非取决于局部剪应力，而是受限于溃口顶部下游紧邻处头坎所维持的静摩擦角φ。图18展示了溃口河道水流的简化二维示意图。自溃口顶部（平均高程z_c）倾泻而下的水流将侵蚀.....

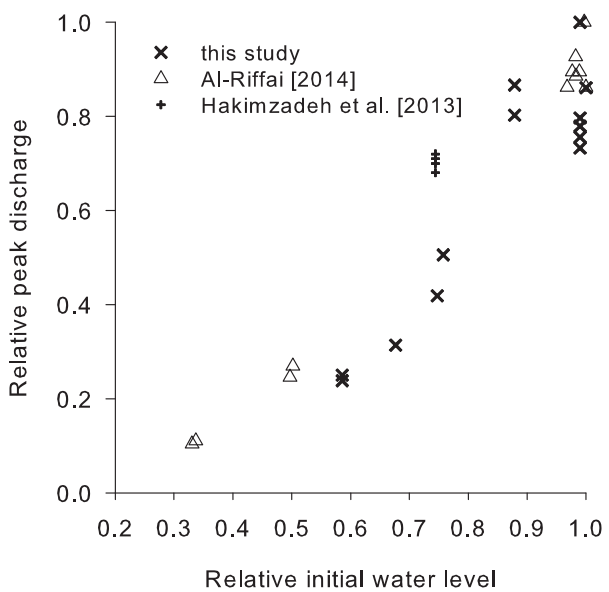


图17. 无量纲洪峰流量随无量纲初始水位变化的关系图，数据来源包括本研究、Al-Riffai[2014]及Hakimzadeh等[2013]。每组数据中，水位标尺取各自实验所用最大值；洪峰流量标尺取该组数据中最高坝体对应的最大值。所有数据集均显示：初始水位降低时，洪峰流量随之减小。

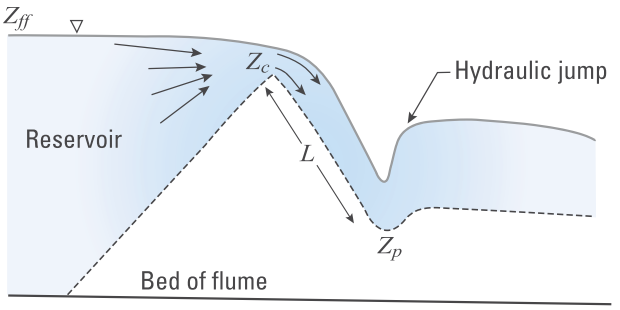


Figure 18. Sketch showing idealized 2-D flow geometry along the centerline of the breach channel. Water level in the far field is z_{ff}. The breach crest is at an average elevation z_c. A headcut at approximately the static angle of friction extends from the breach crest to a plunge pool at elevation z_p; over time the plunge pool is lowered to the bed as scouring proceeds.

headcut; the only way that the slope angle can be maintained at ϕ is if the erosion rate is the same all along the headcut. That erosion rate is set by the sediment flux carried by the flow at the base of the headcut. Empirical equations for total sediment-carrying capacity in sand-bed streams are diverse [*García*, 2008], but all include some measure of the mean flow speed either explicitly or implicitly. To a good approximation [cf. *Chanson*, 2004a, pp. 406–408], Bernoulli's principle (conservation of energy) then tells us that the flow speed at z_p varies as $\sqrt{z_{ff}-z_p}$. If q_s is the sediment-carrying capacity per unit width, then $\dot{\varepsilon}$, the rate of slope-normal erosion of the headcut, will be proportional to q_s/L, where $L=(z_c-z_p)/\sin\phi$ is the length of the headcut. Finally, then, the rate of vertical erosion of the breach crest, given by $|dz_c/dt|=[\sin\theta_u/\sin(\theta_u+\phi)]\dot{\varepsilon}$, is a function of z_{ff}, z_c, z_p, ϕ, θ_u and t. (An argument along these lines has been presented by *Visser* [1994].) All three of z_{ff}, z_c, and z_p will scale with Z_0 as long as erosion proceeds to the bed, in which case Q_p must be a function of Z_0.

6. Summary and Conclusions

A key conclusion of our study is that dam-breach flood hydrographs are quite reproducible if the time datum is chosen as t_{hc}, the time that the migrating headcut formed early in the breaching process intersects the section of hydraulic control, commonly called the breach crest. Hydrograph reproducibility was demonstrated for dams with diverse initial heights. The variability among flood hydrographs during the interval between the start of breaching and t_{hc} is probably a reflection of fairly subtle experiment-to-experiment differences in the groundwater hydrology and the interaction between the flood flow and groundwater.

The migrating headcut that triggers rapid downcutting of the breach crest, and thus rapid increase in flood discharge, is the end stage of an instability that begins when incipiently channelized flow along the downstream dam face becomes supercritical [*Parker and Izumi*, 2000]. The initially uniformly sloped streambed spontaneously transforms into alternating zones of relatively high and low gradient. Eventually channel steps form, steepen, and migrate upstream. The slope of the steep zones is limited by the static angle of friction of the sediment that constitutes the dam. The steep sections eventually merge to form a prominent headcut, downstream of which the breach channel has a relatively gentle slope, with the two sections being separated by a hydraulic jump. There is a suggestion from experiments with broad-crested dams (both ours and those of *Das* [1997]) that the headcut-formation process weakens and eventually vanishes as the ratio of dam-crest breadth to the overall (toe-to-toe) length of the dam increases.

As long as flow is critical at the breach crest and supercritical immediately downstream—and this is the situation except during the earliest stages of the breaching process—the slope failures that shape the breach channel have no influence on the breach crest except for rare cases when the slope failures actually intersect the breach crest. Although there could be, in principle, backwater effects extending as far as the breach crest on account of temporary impoundment behind slump blocks, we never observed this phenomenon. It is therefore incorrect to envisage the geometry of the hydraulic-control section as reflecting a tradeoff between sediment delivery by slope failures and sediment removal by fluvial processes. To the best of our knowledge, all parametric models of breaching incorporate this erroneous concept, which seems to derive from envisaging the breach channel as a gutter and hydraulic control as analogous to a broad-crested weir. In fact, the hydraulic-control section is a purely erosional feature.

The flow cross section at the breach crest evolves in such a way that when all lengths are normalized by the arc length of the crest, the cross section is practically constant through time, at least for noncohesive sediments. The shape of the breach crest is also nearly independent of sediment grain size. Owing to flow

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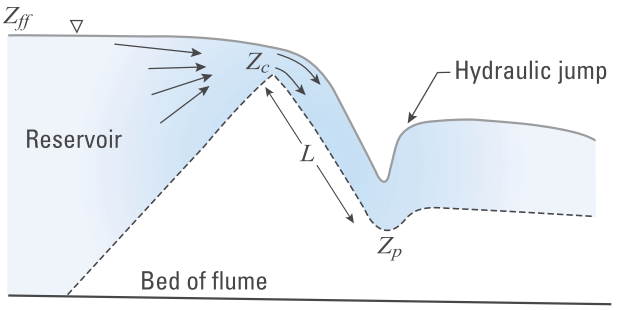


图18. 示意图：沿溃口河道中心线的理想化二维水流几何构型。远场水位为z_{ff}，溃口顶部平均高程为z_c；一近似呈静摩擦角φ的头坎自溃口顶部延伸至高程为z_p的跌水潭；随冲刷持续进行，跌水潭高程逐渐降低至基岩面。

头坎；维持坡角恒为φ的唯一方式是头坎沿程侵蚀速率处处一致。该侵蚀速率由头坎底部水流所携带的泥沙通量决定。砂质河床河流总输沙能力的经验公式多种多样[García, 2008]，但均显式或隐式地包含平均流速指标。依据伯努利原理（能量守恒），可较好地近似得出[z_p处流速随……变化[参见Chanson, 2004a, 第406–408页]。

p。若q_s为输沙能力，单位宽度输沙能力，则头坎沿坡向的侵蚀速率_e与q_s/L成正比，其中L = [z_c − z_p]/sinφ为头坎长度。最终，溃口顶部垂向侵蚀速率dz_c/dt = −e sinφ/sin(φ − θ)为z_c、z_p、φ、θ及t的函数。（Visser[1994]曾提出类似论证。）只要侵蚀持续至基岩面，z_{ff}、z_c和z_p三者均随Z₀按比例缩放，此时Q_p必为Z₀的函数。

6. 总结与结论

本研究一项关键结论是：若以t_{hc}（即溃口初期形成的迁移性头坎与水力控制断面——通常称为溃口顶部——相交的时刻）为时间零点，则溃坝洪水过程线具有高度可重复性。该可重复性已在不同初始高度的坝体实验中得到验证。溃口开始形成至t_{hc}时段内洪水过程线的变异性，可能反映了各次实验间地下水水文条件以及洪水流与地下水相互作用的细微差异。

引发溃口顶部快速下切、从而导致洪峰流量迅速增大的迁移性头坎，是始于下游坝面初具河道化水流转变为超临界流时所触发失稳过程的最终阶段[Parker and Izumi, 2000]。初始呈均匀坡度的河床自发演变为相对高、低坡度交替出现的区段。最终形成阶梯状河床，阶梯不断变陡并向上游迁移。陡坡区的坡度受构成坝体的沉积物静摩擦角限制。陡坡区最终合并形成显著的头坎，其下游溃口河道坡度相对平缓，两区之间以水跃分隔。我们在宽顶坝实验（包括我们自身及Das[1997]的实验）中发现，随着坝顶宽度与坝体总长（坝踵至坝踵）之比增大，头坎形成过程趋于减弱并最终消失。

只要溃口顶部处于临界流状态、且紧邻其下游为超临界流——除溃口初始形成阶段外，此即为典型情形——则塑造溃口河道的边坡失稳现象除极少数情况下实际切割到溃口顶部外，对其不产生任何影响。理论上，滑塌块体后方临时壅水可能引起回水效应并向上游延伸至溃口顶部，但我们从未观测到这一现象。因此，将水力控制断面的几何形态理解为边坡失稳输沙与河流冲刷排沙之间权衡的结果，是不正确的。据我们所知，目前所有参数化溃口模型均纳入了这一错误概念，该概念似乎源于将溃口河道类比为排水沟、并将水力控制类比为宽顶堰。事实上，水力控制断面纯属侵蚀成因。

溃口顶部处的过流断面演化方式为：当所有长度均以溃口顶部弧长为基准进行无量纲化后，该断面在时间上基本保持恒定，至少对无黏性沉积物而言如此。溃口顶部的形态也几乎与沉积物颗粒粒径无关。由于水流作用……

convergence and streamline curvature at the breach crest, the Froude number at the breach crest is about 0.75, not 1, and the ratio of critical depth to specific energy is about 0.8, not 2/3. Nonetheless, the discharge differs only slightly from the value expected for an ideal MEL weir [*Chanson*, 2004a], which has a planform shape very similar to that of the breach crest.

In the proximal breach channel, neither bedload transport nor suspended load transport is in equilibrium with prevailing flow conditions at laboratory scale, although equilibrium might be achieved if the flow path were long enough—that is, at field scale. The distal breach channel is another matter. As streamflow enters the hydraulic jump, the water becomes highly turbid, presumably because both bedload and near-bed suspended load are swept up into suspension by vigorous, chaotic fluid motions. Flow in the distal breach channel is highly unsteady and nonuniform. Furthermore, bank failures in this reach add greatly to the sediment load, in a time and space-dependent fashion. Small slump blocks are commonly swept away more-or-less intact; large blocks become lodged in the breach channel and are then gradually eroded (as previously described by *Morris et al*. [2007] and *Morris and Hassan* [2009]). Given all these complications, it seems unlikely that equilibrium sediment-transport concepts are relevant within the distal breach channel at laboratory scale.

Peak discharge is an increasing function of initial water level. This trend is inconsistent with the seemingly plausible assumption that erosion rate at the breach crest depends upon the local shear stress, but may be understood as a consequence of the hydraulics of flow along a steeply dipping headcut.

Finally, it bears emphasizing that the geometry and constraints in a dam-breach experiment must be carefully considered. All of our experiments were done with what was effectively a specified reservoir volume. Experiments with the reservoir water level held constant [e.g., *Fujita and Tamura*, 1987] are appropriate for modeling breaching of river levees; alternatively, they may be thought of as modeling the drainage of an effectively infinite reservoir. Two-dimensional experiments, in which a dam crest is uniformly overtopped from flume wall to flume wall [e.g., *Schmocker and Hager*, 2009, 2010, 2012], may also be thought of as representing levee failure processes far from the ends of a finite-sized breach. Two-dimensional experiments are also very useful for learning about sediment erosion mechanics. If one wishes to investigate breaching failure of a finite-sized reservoir, however, the challenges of a three-dimensional experiment must be accepted.

Notation

A	cross-sectional area of flow at the breach crest
A_0	initial surface area of reservoir
$\hat{A}$	fictitious A assuming no drawdown at the breach crest
b	bottom width of trapezoidal breach channel in some models
b_c	curve that represents the location of the breach crest
$\tilde{b}$	breadth of dam crest in the average flow direction
B	arc length of breach crest
$\tilde{B}$	straight-line distance between ends of breach crest
c_D	discharge coefficient in weir equation
d	local water depth in hydraulic-control section
d_c	critical depth
D_X	grain size such that X% by weight is finer
E	specific energy (total hydraulic head) at the breach crest
$E_{\max}$	maximum value of E along breach crest
$\hat{E}$	$z_{ff} - z_b$, equal to E in the absence of streamline curvature effects
Fr	Froude number
g	acceleration of gravity
H_d	elevation of dam crest relative to downstream toe of dam
H_u	elevation of dam crest relative to upstream toe of dam
K_{sat}	saturated hydraulic conductivity
L	length of eroding angle-of-repose slope

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溃口顶部的汇流与流线曲率导致其弗劳德数约为0.75，而非1；临界水深与比能之比约为0.8，而非2/3。尽管如此，泄流量与理想MEL堰[Chanson, 2004a]所预期的值仅略有差异，而该理想MEL堰的平面形状与溃口顶部极为相似。

在近端决口通道内，推移质输运与悬移质输运在实验室尺度下均未达到与当前水流条件的平衡状态；尽管若流程足够长——即在野外尺度下——可能实现平衡。远端决口通道的情况则不同。当径流进入水跃区时，水体变得高度浑浊，原因很可能是推移质及近床悬移质在剧烈、紊乱的流体运动作用下被卷入悬浮状态。远端决口通道内的水流具有高度非恒定性与非均匀性。此外，该河段的岸坡失稳以时空依赖的方式显著增加了泥沙负荷。小型滑塌块体通常近乎完整地被冲走；而大型块体则滞留在决口通道内，并逐渐被侵蚀（此前 Morris 等 [2007] 及 Morris 和 Hassan [2009] 已对此类现象作过描述）。鉴于上述种种复杂因素，在实验室尺度下，远端决口通道内应用平衡泥沙输运概念似乎并不适宜。

峰值泄流量随初始水位升高而增大。这一趋势看似与“决口 crest 处侵蚀速率取决于当地剪应力”的合理假定相悖，但实则可由陡倾角溯源侵蚀面水流的水力学特性予以解释。

最后需强调的是，坝体决口实验的几何构型与约束条件必须审慎考量。本研究所有实验均采用实质上固定水库容积的方式进行。而保持水库水位恒定的实验（例如 Fujita 和 Tamura [1987]）适用于模拟河道堤防溃决；或者可理解为模拟一个有效无限大水库的排水过程。二维实验（即坝顶从水槽一侧壁到另一侧壁均匀漫顶，例如 Schmocker 和 Hager [2009, 2010, 2012]）亦可视为模拟有限尺寸决口两端以外区域的堤防失稳过程。二维实验对于认识泥沙侵蚀机理亦极具价值。然而，若旨在探究有限尺寸水库的溃坝失稳过程，则必须接受三维实验所带来的诸多挑战。

符号说明

A 决口 crest 处的过水断面面积；A0 水库初始水面面积

$\hat{A}$ 假设决口 crest 处无跌水时的虚拟 A 值

某些模型中梯形决口通道的底宽 bc

代表决口 crest 位置的曲线

$\tilde{b}$ 坝顶在平均流向上的宽度

B 决口 crest 的弧长

$\tilde{B}$ 决口 crest 两端点之间的直线距离

c_D 堰流公式中的流量系数

d 水力控制断面处的局部水深 dc

临界水深

D_X DX 表示重量占比 X% 的颗粒更细的粒径

E 决口 crest 处的比能（总水力水头）；Emax 沿决口 crest 的 E 最大值

$\hat{}$ zff2zb，在忽略流线曲率效应时等于 E

弗劳德数

g 重力加速度 Hd

坝顶相对于坝体下游趾部的高程

H_u 坝顶相对于坝体上游趾部的高程

Ksat 饱和渗透系数

L 侵蚀坡面（按休止角形成）长度

L_b	toe-to-toe dimension of dam, equal to 3.5 m in all experiments
n	porosity
p	gauge pore pressure
q_s	sediment carrying capacity per unit width
Q	discharge
Q_{in}	recharge
Q_{peak}	peak discharge
s	arc length along the breach crest
$\tilde{s}$	$s/B-1/2$, a dimensionless measure of arc length
t	time
t_p	time to peak discharge
u_*	shear velocity
V_R	reservoir volume
V_0	initial reservoir volume
w_s	particle settling speed
x, y, z	components of position vector relative to locally defined coordinate system
z_b	elevation of points along the breach crest
z_c	$\langle z_b \rangle$
z_d	elevation of the dam face
z_{ff}	reservoir level in the far field, beyond the effect of drawdown near the breach crest
z_p	elevation of foot of the angle-of-repose slope
z_s	elevation of the water surface
z_w	elevation of points on the water surface following the trace of the breach crest
Z_0	initial reservoir level
α	slope of sides of trapezoidal breach channel in some models
δs_1, δs_2	small increments in arc length
Δ	an interval, for example, Δt
$\dot{\varepsilon}$	erosion rate normal to the retreating headcut
θ_d	slope of downstream dam face
θ_u	slope of upstream dam face
θ_r	slope of eroding headcut, commonly close to ϕ
λ	scale factor
ρ_d	dry bulk density of sediment
ρ_s	density of sediment grains
ρ_w	density of water
τ	boundary shear stress
ϕ	angle of static friction
ψ	pressure head
ω	water content expressed as mass water/mass sediment
$\langle\ \rangle$	denotes mean value of quantity between the brackets

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The following are found in the supporting information for this paper: tabulations of material-property measurements; movies cited above; graphs that complement figures presented above; and data that support the various figures. Additional video documentation of our experiments is included in *Logan and Iverson* [2007, revised 2013] and viewable on-line at http://pubs.usgs.gov/of/2007/1315/. We were assisted at the flume by Liz Safran (Lewis and Clark College), Tracy Brennand and Jonathan Cripps (Simon Fraser University), and USGS colleagues Kelly Swinford, Julie Griswold, Rick LaHusen, Steve Schilling, Chris Lockett, Joel Smith, and Roger Denlinger. Schilling also provided help with ArcInfo. National Association of Geology Teachers interns Kate Potter and Frances Griswold helped tremendously with the 2012 and 2013 experiments, respectively. We thank Robert S. Anderson and three anonymous referees for their careful reviews. J. S. W. thanks Mark Morris, Bryan Orendorff, and Mahmoud Al-Riffai for valuable discussions and copies of unpublished data and videotapes; and the late Stephen Coleman for a copy of the thesis by Darryl Andrews. He also acknowledges financial support from a Fulbright grant (U.S. Department of State and Commission franco-américaine) and thanks his hosts at the Laboratoire Magmas et Volcans, Université Blaise Pascal, Clermont-Ferrand, France, where he was based while doing part of the data analysis and writing an early draft. Mention of trade names is for identification purposes only and does not constitute endorsement by the U.S. Geological Survey.

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L_b 坝体趾部至趾部尺寸，在全部实验中均为 3.5 米
n 孔隙率
p 孔隙压力（表压）qs
单位宽度输沙能力
Q 泄流量
Q_{in} 补给
Qpeak 峰值泄流量
s 沿决口 crest 的弧长
~ s=B21=2，无量纲的弧长度量
t 时间
t_p 峰值流量出现时间
u_* 剪切流速
V_R 水库库容
V_0 初始水库库容
w_s 颗粒沉降速度
x；y；z 为相对于局部定义坐标系 zb 的位置矢量分量
溃口 crest 上各点高程
z_c $\langle z_b \rangle$
z_d 坝面高程
z_{ff} 远场水库水位（即溃口 crest 附近因泄流引起的水位降落影响范围之外）
z_p 休止角斜坡坡脚高程
z_s 水面高程
z_w 沿溃口 crest 轮廓线的水面点高程
Z_0 初始水库水位
a 梯形溃口通道侧坡坡度（某些模型中）；ds1、ds2 弧长的微小增量
D 一个区间，例如 Dt
$\dot{\varepsilon}$ 垂直于后退式侵蚀前锋的侵蚀速率
θ_d 下游坝面坡度
θ_u 上游坝面坡度
θ_r 侵蚀前锋坡度，通常接近 /
k 比例系数 qd
沉积物的干容重
ρ_s 沉积物颗粒的密度
ρ_w 水的密度
s 边界剪切应力 / 静摩擦角

w 压力水头
x 含水量，以水质量/沉积物质量表示
你好 表示括号内物理量的平均值

致谢
本文的支撑信息中包含以下内容：材料物性测量数据汇总表；上文所引述的视频资料；对前述图件起补充作用的图表；以及支撑各图件的数据。我们实验的额外视频记录见 Logan 和 Iverson [2007，2013 年修订版]，并可在线查阅：http://pubs.usgs.gov/of/2007/1315/。在水槽实验中，我们得到了刘易斯与克拉克学院 Liz Safran、西蒙弗雷泽大学 Tracy Brennand 和 Jonathan Cripps，以及美国地质调查局同事 Kelly Swinford、Julie Griswold、Rick LaHusen、Steve Schilling、Chris Lockett、Joel Smith 和 Roger Denlinger 的协助。Schilling 还为 ArcInfo 软件使用提供了帮助。美国国家地质教师协会实习生 Kate Potter 和 Frances Griswold 分别为 2012 年和 2013 年的实验提供了巨大帮助。我们感谢 Robert S. Anderson 及三位匿名审稿人的细致评审。

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